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The Conditioning Set Decides the Measured Transfer Asymmetry: An Exact Correction and a Source-Effect Reading for CNN versus Vision Transformer Robustness Comparisons

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ABSTRACT CNN-versus-ViT transfer studies state which samples they score, but those we review never vary this conditioning set. We measure its weight on adversarially trained ResNet-18 and ViT-Tiny (three runs each, full CIFAR-10 test set, per-sample logging). With models, data and attack fixed, four conditioning protocols place the CNN-to-ViT transfer asymmetry between 4.4 and 19.4 points, whereas retraining moves it by at most 2.4 points (run-to-run SD 0.23 to 1.27). The unconditioned estimate lies 9.21 points above the target-correct one, where the law of total probability places it; the premise p, that clean-misclassified samples stay misclassified, is 0.989 to 1.000 on our adversarially trained targets but falls to 0.88 on standard-trained ones, where assuming $p = 1$ errs by up to 1.6 points. The two protocols we endorse differ by 3.91 points (run-level 95% interval [3.12, 4.70]), and successful-source scoring, a different estimand, lies 11.10 points above both-correct. The direction is a source effect of this ViT-Tiny: additive fits to target-correct matrices give source terms of $+5.13$ (CIFAR-10) and $+9.54$ points (CIFAR-100) against target terms of -2.03 and -6.19. The identity term follows the clean-accuracy gap: the three-protocol spread is 1.49 and 1.30 points on near-matched SVHN, 0.19 for a standard-trained pair 0.47 points apart, and rises 0.87 points per point of gap over 28 pool pairs, whereas the successful-source excess does not shrink (9.03 points for two robust WideResNets 0.21 points apart). Checkpoint selection moves the spread by about two points. Three further readings fail direct measurement; we close with reporting requirements.

INDEX TERMS Adversarial robustness, convolutional neural networks, evaluation methodology, deep learning security, reproducibility, transfer attacks, Vision Transformer

I. INTRODUCTION

Deep neural networks are vulnerable to adversarial examples, carefully crafted perturbations that cause confident misclassification while remaining bounded by a small norm budget [1], [2]. As Vision Transformers (ViTs) [3], [4] joined convolutional neural networks (CNNs) [5] as a dominant architecture family, a natural question followed: which family is more robust, and why?

The literature answers this question inconsistently: Bai et al. [6] report that under matched training recipes transformers do not surpass CNNs, whereas Benz et al. [7] report ViTs to be more robust for standardly trained models. Transfer studies add a disagreement of their own: several report that attacks move more easily from ViTs to CNNs than the reverse [7]–[9], whereas Naseer et al. [10] find that perturbations crafted on ViTs transfer poorly under conventional attacks, and the direction measured in this paper is the opposite of the first group's, a point Section V returns to. These disagreements are usually attributed to differences in models, training recipes, or datasets, and the standard response is to compare under a matched recipe.

Yet even when the models, the data, the threat model and the attack budget are all held fixed, a transfer rate still depends on a choice that the studies reviewed in Section II state but do not vary: the *conditioning set*, that is, which samples enter the denominator. The candidates are all test samples, those the target classifies correctly, those both models classify correctly, and those on which the source attack succeeded. Mahmood et al. [8], Ravikumar et al. [11] and Zhao et al. [12] each name the set they use and proceed with it, which is

sound practice, but it leaves open how much a different set would have changed the conclusion on the same models. Robustness comparisons face a parallel choice between an absolute score and its separation into clean accuracy and conditional susceptibility.

The setting is kept small so that this question can be isolated. We train ResNet-18 and ViT-Tiny on CIFAR-10 [13] under one matched recipe, clean pre-training followed by mixed-loss PGD adversarial fine-tuning, with three independent training runs per architecture and a fixed validation split held out before *any* training stage, so that no model is selected on data it has seen. All evaluations run on the full test set with per-sample outcome logging, so the same attack outcomes can be rescored under every protocol without re-running an attack. With the models fixed, only the protocol is varied. Two later additions need no new training: the clean-trained checkpoints of the same pipeline together with a standard-trained WideResNet-28-10, and an eight-model pool that adds robust checkpoints from RobustBench [14].

Architecture comparisons also invite three readings that we treat as measurable hypotheses: that CNN input gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that ViT-specific transfer attacks improve cross-architecture transfer. Only the last is a published claim; the first two are interpretations that a sparsity result or a small batch of attention figures readily suggests, our own earlier figures included.

Our first contribution is a measurement of protocol sensitivity on fixed models. Across four conditioning protocols the CNN-to-ViT transfer asymmetry ranges from $+4.36$ to $+19.37$ points, while within any one protocol retraining both architectures moves it across a range of at most 2.41 points. On CIFAR-10 that range consists of two distances: the successful-source protocol, which estimates a different quantity, lies 11.10 points above the both-correct one, and the two protocols we endorse, target-correct and both-correct, differ by 3.91 points, with a run-level 95% interval of $[3.12, 4.70]$. The unconditioned estimate lies inside the range, 9.21 points above target-correct, where the identity of Section III-F places it once the targets' clean errors, the conditioned rates and the measured premise are known. Whether an equivalence claim can be made at all also depends on the protocol (Section IV-D2).

The second contribution concerns unconditioned rates. Their deviation from the target-correct rate follows from an elementary identity, the transfer analogue of the TRADES decomposition [15], whose premise we measure: the probability p that a sample the target already misclassifies stays misclassified under a transferred perturbation. On adversarially trained targets p lies between 0.989 and 1.000 over 36 directions, so the correction is exact to within a few tenths of a point. On standard-trained targets p falls to 0.877 and the $p = 1$ form misses the unconditioned rate by up to 1.58 points, so there the general form, with p measured, has to be used.

The third contribution locates the direction. In 3×3 transfer matrices that add a WideResNet-28-10, an additive source-plus-target fit to the target-correct rates assigns the asymmetry to the source: examples crafted on our ViT-Tiny transfer worse to every target, the external one included, while the ViT is the *less* vulnerable target (source terms $+5.13$ and $+9.54$ points on CIFAR-10 and CIFAR-100, target terms -2.03 and -6.19). The source term keeps its sign under every conditioned protocol, whereas the target term depends on the protocol and is indistinguishable from zero under both-correct on CIFAR-10, so the direction is best read as a property of the examples this ViT-Tiny produces. Read through raw rates, the same matrices make the ViT look like the weaker target, because raw rates carry its clean error. In the 8×8 matrices of Mahmood et al. [8] the source term favors the ViT instead, and their CIFAR-100 pairs that share our direction reach it through the target term.

The fourth contribution is a matched-pair prediction that follows from the identity and that the data confirm for the identity term only. Because the unconditioned, target-correct and both-correct estimates are separated by terms proportional to clean errors, their spread should scale with the pair's clean-accuracy gap and vanish for pairs matched in clean accuracy, whatever their architectures. What the data add is measured, not derived: p is at least 0.984 on every adversarially trained target we examined, the both-correct estimate lies between the other two in 57 of 60 model pairs, and the successful-source term is unrelated to the gap. The three-protocol spread is accordingly 1.49 and 1.30 points on SVHN, where our two models are 1.84 points apart in clean accuracy, and 0.19 points for a standard-trained ResNet-18 and WideResNet-28-10 that are 0.47 points apart, and across the 28 pairs of the eight-model pool it rises by 0.872 points per point of gap. The successful-source excess does not shrink: two robust WideResNet-28-10 checkpoints 0.21 points apart still have a four-protocol spread of 9.03 points. The checkpoint-selection rule moves the clean gap by about 1.7 points and the four-protocol spread by about two, without changing the direction.

Absolute robustness favors the CNN under every attack and factors exactly into clean accuracy and conditional susceptibility; both factors favor the CNN in the present runs, although their attribution differed on an earlier checkpoint set. Three further readings do not survive direct measurement, and Section IV-G reports them together: adversarially trained CNN gradients are sparser but not more spatially localized, CLS attention entropy does not change under attack, and a ViT-specific transfer attack gives no gain in this regime. The reporting requirements that follow from these measurements are collected in Section V-B.

A preliminary version of this study was submitted to ATEEC 2026 [16]. That version reports the conditional decomposition, the four-protocol transfer comparison, the gradient structure measurements and the block-wise drift profiles on CIFAR-10 alone, and describes them as preliminary. The present paper adds two further datasets, a second threat model, the identity of Section III-F with its measured premise,

the source-target decomposition, standard-trained targets and an external model pool, the analysis of checkpoint selection, the transformer-specific transfer attack and an analysis protocol registered before measurement, and it corrects one claim of the earlier version: the direction of the asymmetry is not stable once the architectures are matched in clean accuracy.

After a review of adversarial evaluation practice and architecture-comparison studies in Section II, Section III describes the training protocol, the conditioning protocols and the measurements, and Section IV reports the results. Implications and limitations follow in Section V, and Section VI concludes.

II. RELATED WORK

A. ADVERSARIAL ATTACKS

Adversarial attacks search for small perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] takes a single step along the sign of the loss gradient within an L_∞ budget, and Projected Gradient Descent (PGD) [17] iterates that step with a projection back onto the ϵ-ball (Section III-C). The Carlini & Wagner attack [18] casts the search as an optimization problem, and AutoAttack [19], an ensemble of APGD-CE, APGD-T, FAB-T and Square Attack, has become the de facto standard for reliable robustness evaluation.

B. DEFENSE MECHANISMS

Adversarial training [17], which minimizes the loss on worst-case perturbations generated during training, remains the most effective empirical defense. TRADES [15] splits the robust error into a natural and a boundary term and trades the two off explicitly, and MART [20] gives misclassified examples more weight during training. Other defenses transform the input [21], certify robustness by randomized smoothing [22], which Adaptive Randomized Smoothing [23] extends to test-time adaptive, multi-step models, or rely on ensembles [24].

C. CNN ROBUSTNESS

For CNNs, Madry et al. [17] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [14] tracks robust accuracy across architectures, and WideResNet variants occupy the top of its CIFAR-10 ranking.

Studies have shown that model capacity correlates with achievable robustness [17], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the role of skip connections in adversarial transferability [25] and the distinct batch-normalization statistics that adversarial training induces at scale [26]; relatedly, auxiliary batch-normalization branches allow adversarial examples to improve clean accuracy [27]. Adversarial training is also known to reshape input-gradient structure itself, yielding more perceptually aligned gradients [28] and sparser feature attributions [29].

D. VISION TRANSFORMER ROBUSTNESS

Vision Transformers [3], building on the self-attention mechanism introduced by Vaswani et al. [4], process images as sequences of patches; their internal representations differ measurably from CNNs' layer by layer [30]. Bhojanapalli et al. [31] found ViTs at least as robust as ResNets across a broad range of perturbations when pre-trained on enough data. Bai et al. [6] systematically investigated whether transformers are more robust than CNNs, finding that under fair training recipes transformers are not more adversarially robust than CNNs, while exhibiting superior robustness under distribution shift attributed to self-attention; Benz et al. [7] reported an early direct comparison including MLP-Mixer and linked the differences to frequency preferences. Subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [8], with patch-targeted attacks exploiting the attention mechanism directly [32] and attention shown to shift measurably under patch perturbations [33].

Shao et al. [9] found ViTs more robust than CNNs and MLP-Mixers in white-box and transfer settings, with features that carry less high-frequency content, and showed that adversarial training applies to ViTs, although in their preliminary CIFAR-10 run the adversarially trained ViT did not improve on a large CNN. ViT robustness to common corruptions and occlusions has also been characterized [34]. More recently, Jain et al. [35] traced a form of gradient masking in ViTs to softmax underflow in attention, which leaves PGD-style attacks sub-optimal on them, a caveat for PGD-only ViT evaluations that AutoAttack, with its gradient-free Square component, partly guards against; the adversarial-training recipe they build on the corrected attack improves on existing ViT recipes on CIFAR-10 and ImageNet-100; with diffusion-augmented adversarial training, ViTs have even been reported to surpass WideResNets on small-scale datasets [36]. Zhu et al. [37] strengthened transferable attacks on ViTs through gradient normalization scaling and high-frequency adaptation. Behavioral comparisons between CNN and ViT robustness under a single evaluation protocol remain scarce.

E. TRANSFER ATTACKS

Adversarial examples often transfer between models, enabling black-box attacks [38]. Demontis et al. [39] trace transferability to two factors, the intrinsic vulnerability of the target model and the complexity of the surrogate, and connect it to the alignment of the two models' input gradients. Foundational transfer studies already evaluate transferability on correctly classified samples [40], and standard attack papers condition their success rates for the same reason [41]; systematic re-evaluations of transfer attacks likewise restrict the pool to samples that every model classifies correctly [12]. Two recent benchmarks move in the same direction: TA-Bench [42] standardizes the comparison of more than thirty transfer attacks over a common pool of substitute and victim models, and Yu et al. [43] report that transferability is systematically overestimated when it is measured across a single

architecture family, and propose three evaluation protocols that differ in how a multi-target outcome is scored: treating an ensemble as a single target, counting how many targets are fooled, and requiring that all of them are. TA-Bench holds the scoring rule fixed and varies the attack and the model pool; Yu et al. vary the scoring rule along the target-aggregation axis. We vary it along the conditioning axis instead, with the models, the data and the attack budget fixed, which isolates the contribution of the conditioning set and makes it possible to state the arithmetic relation between the resulting estimates. In ensemble design, Ravikumar et al. [11] restrict transfer measurement to samples correctly classified by both models precisely to account for accuracy differences between them, the same confound our conditioned metric controls.

Mahmood et al. [8] measured transfer between CNNs and ViTs with a both-correct protocol and found notably low CNN↔ViT transferability; Naseer et al. [10] showed that adversarial examples crafted *on* ViTs transfer poorly under conventional attacks and proposed self-ensemble and token refinement to improve ViT-sourced transferability, and Wei et al. [44] improved ViT-sourced transfer by bypassing attention gradients and patch-wise dropout. Building on these, Token Gradient Regularization (TGR) [45] crafts transferable adversarial samples by exploiting ViT's attention components, while Momentum Integrated Gradients (MIG) [46] achieves improved transferability across both ViTs and CNNs. Chen et al. [47] propose a transfer attack that exploits the differing inductive biases of CNNs and ViTs; the frequency preferences behind that difference, CNNs leaning on high-frequency and ViTs on low-frequency content, are documented by Benz et al. [7] and Shao et al. [9]. Waseda et al. [48] separate transfer outcomes further, by whether the target reproduces the source's exact error or fails in a different way, and report that both occur even between closely related models.

F. RESEARCH GAP

While individual studies have examined CNN and ViT robustness separately, and the pitfalls of unfair CNN and transformer comparisons have been highlighted before [6], [49], existing comparative work typically reports robustness scores in isolation, often with inconsistent perturbation budgets, attack configurations, or model scales, and rarely pairs the comparison with behavioral analysis. A recent study in this journal compares ViT and CNN adversarial robustness directly [50]; our work differs by evaluating on the full test set with AutoAttack and paired statistics, by controlling transfer measurement for clean-accuracy differences, and by accompanying the comparison with gradient- and feature-level behavioral analyses.

The evaluation-methodology literature has repeatedly shown that how robustness is measured decides what is concluded: Carlini et al. [51] catalog the failure modes of robustness evaluation, and Tramèr et al. [52] show that defenses judged robust under fixed attacks fall to adaptive ones. That work concerns attack strength, that is, whether the measurement is strong enough. The question we ask is orthogonal: with attack strength held fixed, how much does the *conditioning set* alone move the conclusion? Conditioning transfer evaluation on samples the models classify correctly is established practice [8], [11], [40], [41], and each of these studies states its conditioning set and proceeds with it; Yu et al. vary the scoring rule, but along a different axis. None of the studies cited here varies the conditioning set on fixed models, which is the design we use. The confound in unconditioned rates then follows from the law of total probability: when samples the target misclassifies stay misclassified, $r_{\text{raw}} - r_{\text{cond}} = e\,(1 - r_{\text{cond}})$, the transfer analogue of the natural-plus-boundary decomposition of TRADES [15]. In the transfer setting the premise of that decomposition is an empirical quantity, which we measure, and the relation turns a known qualitative caveat into a correction that applies to published raw rates wherever the premise holds.

The same design lets us test three readings that architecture comparisons tend to suggest: that CNN gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that transformer-specific transfer attacks [10], [45] improve cross-architecture transfer. Only the third is claimed in the cited work; the first two are interpretations. We measure each directly, and none holds in our setting.

III. METHODOLOGY

Figure 1 summarizes the design. Two architectures are trained under one matched adversarial recipe with three independent runs each, and every evaluation is logged per sample on the full test set, so that, within each analysis, the four conditioning protocols of Section III-F can be applied to the same records without re-running the attack. The subsections below fix the threat model, the models, the attacks and the training protocol in that order, and then define the measurements.

A. THREAT MODEL

We consider the standard L_∞ threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(x) = \{x' : \|x' - x\|_\infty \leq \epsilon\} \tag{1}$$

Following established conventions for CIFAR-10 [13], [14], we use $\epsilon = 8/255 \approx 0.031$ as the primary perturbation budget, with additional full-test PGD-10 evaluations at $\epsilon \in \{2/255, 4/255, 16/255\}$ for the epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

B. MODEL ARCHITECTURES

a: CNN Models

ResNet-18 [5] is the CIFAR-10 variant of the residual network, with an initial 3×3 convolution in place of the 7×7 stem and 11.2M parameters. WideResNet-28-10 [53], with

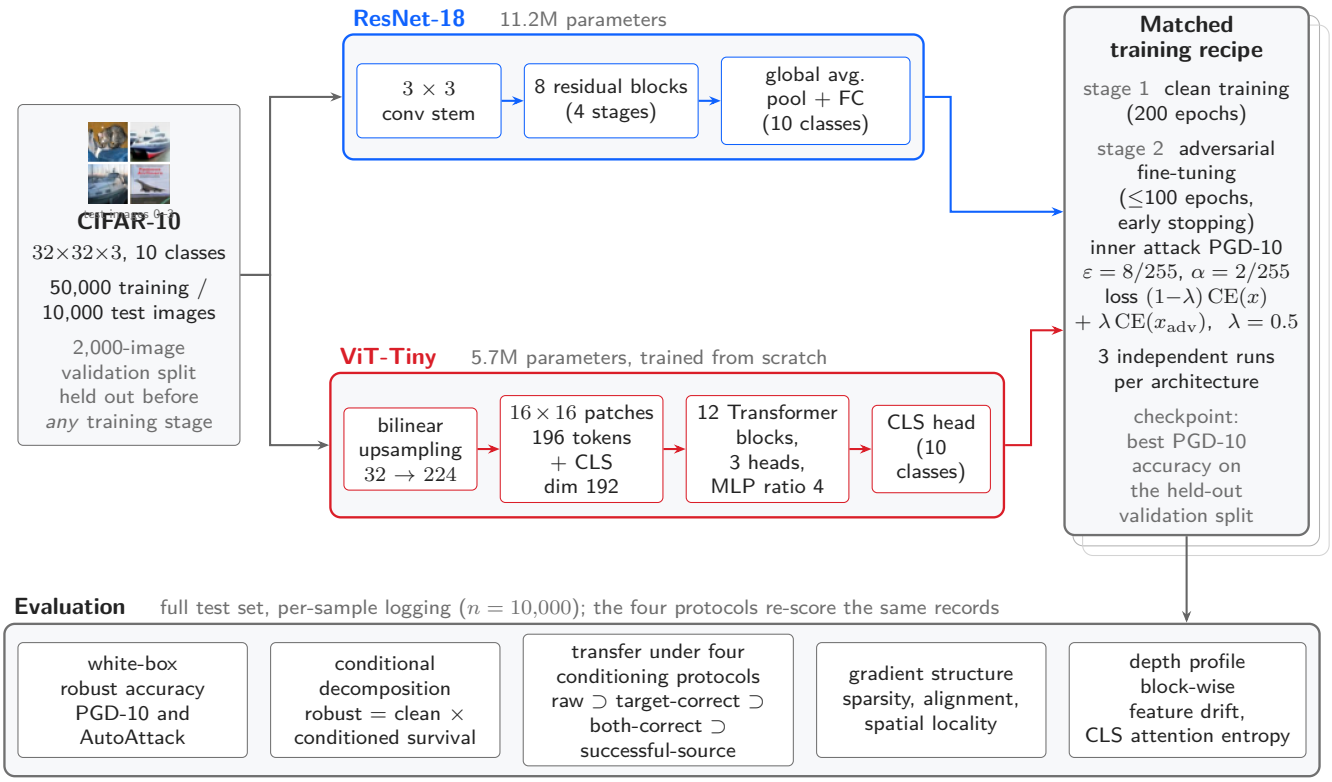


FIGURE 1. Study overview. The same CIFAR-10 split feeds two architectures, ResNet-18 and ViT-Tiny, which are trained under one matched adversarial recipe with three independent runs each and selected on a validation split held out before any training. Every evaluation is logged per sample on the full test set, which is what allows the four conditioning protocols to be applied to identical records. The five measurement families in the bottom row are the ones Section IV reports; the strip beneath them lists the replication settings and controls.

36.5M parameters, is the usual reference architecture for adversarial robustness on CIFAR-10; we use the robust checkpoint of Gowal et al. [54] distributed through RobustBench [14]. It was trained with additional unlabeled data and a stronger recipe than ours, so it serves as a reference point and not as a matched comparison. Section IV-E adds further RobustBench checkpoints, a standard-trained WideResNet-28-10 among them, without new training.

b: Vision Transformer Models

ViT-Tiny comes from the `timm` library [55] and is adapted for CIFAR-10 by bilinearly upsampling inputs to 224×224. The architecture uses 16×16 patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. The model is trained *from scratch* on CIFAR-10 (no ImageNet pretraining) with basic augmentation (random crop and horizontal flip), then adversarially fine-tuned. Since ViT robustness on small datasets is known to be sensitive to the training recipe [56], [57], our architecture-level conclusions should be interpreted as holding *under this baseline training recipe* and not for optimally trained ViTs.

C. ATTACK METHODS

a: FGSM

Single-step attack with step size equal to the perturbation budget:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{CE}(f(x), y)) \tag{2}$$

b: PGD

Iterative attack with step size $\alpha = 2/255$ and random initialization:

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)}\left(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}_{CE}(f(x^t), y))\right) \tag{3}$$

with $x^0 = x + \mathcal{U}(-\epsilon, \epsilon)$. We use PGD-10 (10 iterations) for both adversarial training and evaluation throughout this work (Table 2).

c: AutoAttack

We use AutoAttack [19], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-T (Auto-PGD with targeted DLR loss), FAB-T (targeted Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack).

D. DEFENSE METHODS

a: Adversarial Training (AT)

We employ a mixed adversarial training objective that combines clean and adversarial losses:

$$\mathcal{L} = (1-\lambda)\,\mathcal{L}_{CE}(f_\theta(x), y) + \lambda\,\mathcal{L}_{CE}(f_\theta(x_{adv}), y) \quad (4)$$

where $\lambda = 0.5$ balances standard classification performance with adversarial robustness. The inner maximization generates adversarial examples using PGD-10 ($\alpha = 2/255$, $\epsilon = 8/255$), computed with the model in training mode so that attack generation and the parameter update see consistent batch-normalization statistics, as in the reference implementation of Rice et al. [58]. Mixing the two losses helps maintain clean accuracy while building robustness, since training solely on adversarial examples can degrade natural performance. We refer to the recipe as *mixed-loss PGD adversarial fine-tuning*: half of the loss is clean and it is applied to clean pre-trained checkpoints, so it is not the PGD adversarial training of Madry et al. [17], and the robust accuracies it yields should not be read against reference PGD adversarial training results. Alternative formulations such as TRADES [15] offer theoretically grounded trade-off mechanisms; we keep one recipe for both architectures to isolate architectural effects from defense-specific interactions.

E. TRAINING PROTOCOL

a: Clean Training

For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule [59], weight decay 5×10^{-4}, and 200 epochs. For ViTs, we use AdamW optimizer [60] with initial learning rate 10^{-3}, cosine annealing, weight decay 0.05, and 200 epochs.

b: Adversarial Training

Starting from our clean-trained checkpoints, we fine-tune with reduced learning rates: CNNs use SGD with LR 10^{-3} and cosine annealing for up to 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Gradients are clipped to a global L_2 norm of 10, and batches with non-finite loss are skipped. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping; all models consume raw pixel inputs in $[0, 1]$ without mean/std normalization, and attacks and epsilon budgets operate directly in this pixel space. The training batch size is 128 for the CNN and 64 for the ViT (evaluation batch sizes vary by analysis). Early stopping uses patience 20 with a 0.1 percentage-point improvement threshold.

c: Validation Split and Repetitions

Model selection and early stopping use PGD-10 adversarial accuracy on a fixed 2,000-sample validation split. The split is drawn once (seed 777, indices stored with the code) and is shared by every model in the study, and it is excluded from the training set of *both* the clean and the adversarial stage, so no checkpoint is ever selected on data it was trained on. The test set is touched only by the final post-hoc evaluations. We repeat the full pipeline, clean pre-training followed by adversarial fine-tuning, three times per architecture with independent seeds (1001 to 1003 for ResNet-18, 2001 to 2003 for ViT-Tiny), and pair the runs by index so that every cross-architecture comparison is between models trained under the same repetition. Tables report the mean and standard deviation over these three runs except where stated: SVHN has two runs, whose values are printed individually, and the WideResNet-28-10 reference and the RobustBench checkpoints of Section IV-E are single external models.

An earlier pair of checkpoints, trained before this protocol was adopted and selected on a validation split that clean pre-training had already seen, gave the same absolute ordering as the runs reported here but a different division between clean accuracy and conditional susceptibility; because its records predate the leakage-free protocol, they are not released and none of its numbers enters this paper.

F. EVALUATION METRICS

a: Robustness Metrics

We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack). For white-box evaluations we additionally report the *conditioned fooling rate*, the fraction of clean-correct samples that the attack flips, and its complement, conditioned survival. Because both PGD and AutoAttack count clean-misclassified samples as non-robust in our per-sample logs, absolute robust accuracy factorizes exactly as robust = clean $\times$ conditioned survival, the white-box counterpart of the natural-plus-boundary split of TRADES [15], which lets us decompose robustness differences into a clean-accuracy component and a conditional-susceptibility component. Accuracies are also reported on the *both-correct* subset (samples that both compared models classify correctly on clean inputs), which yields an exactly paired comparison on a shared sample set.

b: Transfer Attack Metrics

Our transfer metrics are a family of *conditioned fooling rates* that differ only in the conditioning set. The base member is the *target-correct* rate: among the samples that the target model classifies correctly on clean inputs, the fraction that becomes misclassified under adversarial examples crafted on the source model. Given adversarial examples $\{x^S_{adv}\}$ generated against source model f_S, the conditioned transfer rate to target model f_T is:

$$T_c(f_S \to f_T) = \frac{\sum_{i=1}^{n} \mathbf{1}[f_T(x_i) = y_i]\,\mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]}{\sum_{i=1}^{n} \mathbf{1}[f_T(x_i) = y_i]} \quad (5)$$

Conditioning transfer evaluation on correctly classified samples is established practice, usually on the samples that every model in the study classifies correctly [8], [11],

TABLE 1. Conditioning masks of the four transfer protocols for a source f_S and a target f_T. A sample (x, y) enters the denominator when every listed condition holds; the numerator counts the samples of the mask with $f_T(x^S_{adv}) \neq y$. The last column states whether the two directions of a pair are scored on the same samples. The loose reading of successful-source is a sensitivity analysis only.

Protocol	Sample enters the denominator if	Same set
Unconditioned (raw)	always	yes
Target-correct	$f_T(x) = y$	no
Both-correct	$f_T(x) = y$ and $f_S(x) = y$	yes
Successful-source	$f_T(x) = y, f_S(x) = y, f_S(x^S_{adv}) \neq y$	no
loose reading	$f_T(x) = y, f_S(x^S_{adv}) \neq y$	no

[12], [40], [41]; the target-correct rate is the one-sided relaxation of that pool, and conditioning matters most in cross-architecture comparisons: the raw misclassification rate $\frac{1}{n}\sum_i \mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]$ includes the target's clean error, so a target with lower clean accuracy appears more "vulnerable" to transfer even when no additional samples are flipped by the attack. We report the raw rate as a secondary metric for transparency and to quantify the size of this confound. Because the conditioning set itself is a protocol choice, we additionally report two alternatives: the *both-correct* protocol, which conditions on samples that both the source and the target classify correctly [8], [11], and the *successful-source* protocol, which conditions on target-correct samples whose attack succeeded on the source in the white-box sense. Both are established. The successful-source rate is the transferability metric of Ravikumar et al. [11], whose Equation (1) divides the number of examples that transfer to the target by the number generated against the source, over a pool restricted to samples both models classify correctly; it is also the protocol implied whenever a transfer rate is reported over "successful" adversarial examples. For directional claims we take the both-correct rate as the primary estimate, because it scores both directions on identical samples and therefore supports paired inference; the other three are reported beside it throughout. As Section IV-B shows, these protocols can yield qualitatively different asymmetry conclusions, which is why every transfer number in this paper is given together with its protocol. Transfer rates are computed on the full test set (10,000 samples) with per-sample outcomes logged, and we report 95% confidence intervals obtained by percentile bootstrap over samples (10,000 resamples).

c: Conditioning Masks and the Raw-Rate Identity

Table 1 defines the four protocols in one place, together with the two readings of successful-source. Every protocol counts, in its numerator, the samples of its mask that the target misclassifies on the source-crafted input; the protocols differ only in the mask. The successful-source protocol needs this table most, because its name does not say whether white-box success requires the source to classify the clean input correctly. We use the strict reading throughout and report the loose one only as a sensitivity (Section IV-B).

The unconditioned and target-correct rates are tied together by the law of total probability. Write e for the target's clean error, r_{raw} and r_{cond} for the unconditioned and target-correct rates of a direction, and p for the probability that a sample the target misclassifies on clean input is also misclassified on the source-crafted input. Then

$$r_{\text{raw}} = e\,p + r_{\text{cond}}\,(1 - e), \tag{6}$$

and, when $p = 1$,

$$r_{\text{raw}} - r_{\text{cond}} = e\,(1 - r_{\text{cond}}). \tag{7}$$

Written over a single model, $r_{\text{raw}} = e + r_{\text{cond}}(1 - e)$ is the natural-plus-boundary decomposition of TRADES [15], where the premise corresponding to $p = 1$ holds by definition because the boundary term quantifies over the whole perturbation neighborhood. In the transfer setting the premise is not definitional: a fixed source-crafted perturbation can leave a misclassified sample misclassified or move it back to the correct class, so p is an empirical quantity that has to be measured. Equation (6) is an accounting identity that holds by construction on any set of samples, so its only empirical content is the value of p; the measurement this paper adds is that value. Where $p < 1$, Equation (7) overstates the deviation by exactly $e\,(1 - p)$ and is an upper bound, not an equality. We therefore use Equation (7) as an exact correction only for targets on which p has been measured close to one, which in this paper means the adversarially trained targets of Section IV-B1; on standard-trained targets (Section IV-E) the general form, Equation (6), with p measured, is the correction to use.

d: Gradient Analysis Metrics

We analyze gradient properties through three complementary metric families, computing gradients of per-sample losses so that magnitudes are independent of the evaluation batch size. Gradient norm ($\|\nabla_x \mathcal{L}\|_2$ and $\|\nabla_x \mathcal{L}\|_\infty$) measures overall sensitivity magnitude. Gradient sparsity is quantified with scale-invariant measures: Hoyer sparsity, the Gini coefficient of $|g|$, and the fraction of components below 1% of the per-sample maximum, since fixed absolute thresholds conflate gradient scale with structure; Hoyer and Gini satisfy the scale-invariance axioms formalized by Hurley and Rickard [61]. Gradient alignment measures the average pairwise absolute cosine similarity across samples (computed over all pairs of the 500 analysis samples):

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i} \mathcal{L} \cdot \nabla_{x_j} \mathcal{L}}{\|\nabla_{x_i} \mathcal{L}\| \|\nabla_{x_j} \mathcal{L}\|} \right| \tag{8}$$

We use the absolute cosine because shared sensitivity *axes* matter for attack geometry regardless of sign; since sign coherence is what universal-perturbation arguments additionally require, we also report the signed mean cosine, which turns out to be near zero for every model (Section IV-G1). High absolute alignment indicates similar perturbation axes across inputs.

e: Spatial Locality Metrics

Sparsity measures how many gradient components are large but says nothing about where they lie in the image, so we measure spatial structure separately on the channel-summed energy map $e = \sum_c (\nabla_x \mathcal{L})_c^2$, normalized to sum to one. We report the smallest pixel fraction whose energy sums to 50% and to 90% (smaller means more localized), the Shannon entropy of e viewed as a distribution over pixels (lower means more localized), and Moran's I with 4-neighborhood weights, which is positive when energy clusters in adjacent pixels. All four are invariant to the scale of the gradient, and all are computed per sample so that the two architectures can be compared with paired tests on identical inputs.

f: Token Aggregation for Layer-wise Analysis

A transformer block emits one vector per token, so summarizing a block requires a choice. We report three: the CLS token alone, the mean over the 196 patch tokens, and the concatenation of all tokens flattened per sample. For the CNN each residual block output is flattened over its spatial dimensions. We report all three for the ViT because, as Section IV-G2 shows, they give different drift magnitudes and place the minimum at different depths.

g: Attention Metrics

For the ViT we extract the post-softmax CLS-to-patch attention row of each block, averaged over heads. We report its Shannon entropy, which measures how broadly attention is spread, and its displacement under attack, measured as the total variation distance between the clean and adversarial attention distributions (bounded by 1). Both are computed per sample and averaged, on $n = 1000$ samples per run.

h: ViT-Specific Transfer Attack

To test whether the transfer results depend on using architecture-agnostic attacks, we additionally generate source examples with Token Gradient Regularization (TGR) [45], which zeroes the extreme-valued entries of the token-gradient tensors in the attention, QKV, and MLP paths during backpropagation and scales the remaining gradient by $\gamma = 0.5$. Our implementation adapts the method to our CIFAR-10 ViT wrapper and to the budget used throughout this paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) instead of the original ImageNet defaults. Because the result it produces is negative, two further details of our version matter. We apply a single $\gamma = 0.5$ to all three paths rather than tuning it per path, and we zero one maximum and one minimum entry per (sample, channel) across tokens rather than whole token rows and columns. Because the published regularization scheme is not replicated exactly, the negative result below bounds our adaptation only; extending it to the method as published would require a replication with the original scheme. Plain MI-FGSM [41] examples are generated in the same run on the same samples, giving a paired control at an identical budget.

i: Feature Degradation Metrics

To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer ℓ. Feature cosine similarity measures directional preservation of the representation:

$$\mathrm{Sim}_\ell = \frac{h_\ell(x) \cdot h_\ell(x_{adv})}{\|h_\ell(x)\| \|h_\ell(x_{adv})\|} \tag{9}$$

where h_ℓ denotes the representation at layer ℓ. Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|h_\ell(x_{adv})\| - \|h_\ell(x)\|}{\|h_\ell(x)\|} \times 100\% \tag{10}$$

Feature L_2 distance ($\|h_\ell(x) - h_\ell(x_{adv})\|_2$) quantifies absolute deviation.

G. STATISTICAL VALIDATION

We separate four sources of variance and report each where it applies. Architecture comparisons turn on training-run variance, so every table reports the mean and standard deviation over three independent runs per architecture (Section III-E). Attack-initialization variance is measured by rerunning PGD-10 with three attack seeds (42, 123, 456) on a fixed subset; the evaluated samples and their order are fixed, so the seeds vary only the random start of the attack. Protocol variance, the spread across conditioning protocols, is treated as a result and not as noise (Section IV-B). Checkpoint-selection variance, the fourth, comes from the offline rule that picks the final checkpoint from a fixed trajectory (validation split, early-stopping patience, and curve smoothing), measured over eighteen combinations while the trajectory itself is held constant. Section IV-H compares their magnitudes, and Section IV-F follows the selection rule into the transfer asymmetry.

Paired tests are used wherever the same samples are scored under two conditions: McNemar's exact test for paired binary outcomes, Wilcoxon signed-rank with Holm correction for the three sparsity measures, paired bootstrap confidence intervals (10,000 resamples) for differences, and a sign-flip permutation test (20,000 permutations) for the both-correct transfer difference. Two units of inference are available throughout and they answer different questions. A per-sample test asks whether two fixed checkpoints differ on the images evaluated; a run-level test, over the three repetitions, asks whether the architectures differ. Only the second supports an architectural claim, and with three runs it is much the weaker. The unit is named next to every test, the run level is used wherever the claim is architectural, and where the two units disagree both are reported instead of the one that favors the claim; Section IV-G1 is where that happens. Equivalence claims use two one-sided tests (TOST) at a primary margin of ± 2 points, fixed before the comparisons from the run-to-run variation in PGD accuracy. That is the right scale for an absolute accuracy and the wrong one for the transfer asymmetry, whose run-to-run standard deviation is 0.23 points, so there the margin is some eight times too generous and biases

TABLE 2. Main robustness results on CIFAR-10 ($\epsilon = 8/255$), mean±std over three independent training runs per architecture. AA denotes AutoAttack with the standard suite.

Model	Defense	Params	Clean	FGSM	PGD-10	AA
CNN Models						
ResNet-18	None	11.2M	95.25±0.15	40.29±1.77	0.03±0.03	–
ResNet-18	AT	11.2M	85.78±0.36	53.46±0.08	44.11±0.50	**37.93±0.14**
WRN-28-10	AT†	36.5M	89.48	70.91	66.92	62.76
Vision Transformer Models						
ViT-Tiny	None	5.7M	80.09±0.60	6.10±1.47	0.05±0.05	–
ViT-Tiny	AT	5.7M	73.53±0.55	36.31±0.42	32.69±0.22	**29.14±0.40**

†Robust checkpoint of Gowal et al. [54] via RobustBench [14]; a single external checkpoint, so no standard deviation applies. Clean/FGSM/PGD-10: our local full-test evaluations; AA: the RobustBench-reported figure (we did not rerun AutoAttack on it).

TOST toward accepting equivalence. We keep the margin fixed before measurement instead of retuning it after seeing the data, and report every equivalence claim at ± 1 and ± 3 points as well; the one equivalence we accept for a transfer asymmetry, on SVHN, holds at ± 1 too. The equivalence of attention entropy in Section IV-G2 is a separate test on a different scale, with its own margin of ± 0.05 nats fixed from the scale of that measure. All evaluation and analysis scripts fix random seeds (seed 42, including CUDA determinism flags), and every protocol estimate is recomputed from the stored per-sample records with the masks of Table 1.

IV. EXPERIMENTS

A. ROBUSTNESS COMPARISON

Table 2 gives absolute robustness. Every cell for our models is the mean over three independent training runs per architecture, with the standard deviation over those runs; each run is evaluated on the full 10,000-sample test set.

The CNN holds an *absolute* robustness advantage under every attack: ResNet-18 AT reaches $37.93 \pm 0.14\%$ robust accuracy under AutoAttack against $29.14 \pm 0.40\%$ for ViT-Tiny AT. Because both models of a pair are evaluated on the same 10,000 samples with per-sample outcomes logged, McNemar's paired test applies within each run and favors the CNN in all three pairs ($p < 2 \times 10^{-84}$ under AutoAttack, $p < 2 \times 10^{-120}$ under PGD-10). Seed-level variation for the adversarially trained pair is at most 0.55 points on any reported accuracy, so the ordering is not run-to-run noise. The CNN also retains higher clean accuracy (85.78 vs 73.53%, a 12.3-point gap), reflecting the difficulty of training ViTs on small datasets from scratch, and AutoAttack falls below PGD-10 for both architectures (6.2 points for ResNet-18, 3.6 for ViT-Tiny). The WideResNet-28-10 reference [54], at 62.76%, shows how far capacity, extra training data and a stronger recipe jointly push robustness beyond the architectural choice; we keep it separate from the matched pair throughout.

Table 3 decomposes the absolute scores into their clean-accuracy and conditional-susceptibility components (Section III-F). Because misclassified clean inputs count as non-robust, the factorization is exact: for AutoAttack it reads $85.78 \times 0.442 = 37.93$ for the CNN and $73.53 \times 0.396 = 29.14$ for the ViT. Both factors favor the CNN. It starts from a 12.3-point higher clean accuracy *and* loses a smaller fraction of what it solves, with conditional fooling of 48.58 versus 55.53% under PGD-10 and 55.78 versus 60.37% under AutoAttack. On the $n = 7{,}061 \pm 68$ samples both models classify correctly on clean inputs, an exactly paired comparison, the CNN leads by 13.8 and 10.7 points (McNemar $p < 10^{-67}$ in every run; the bound is computed on this subset and is weaker than the full-test-set bound quoted above, which is what the larger denominator buys).

The decomposition is worth reporting even when the two factors agree, because they need not: an earlier pair of checkpoints, trained before the fixed-split protocol and selected on a validation split that clean pre-training had already seen, gave the same absolute ordering but a different division between clean accuracy and conditional susceptibility (Section III-E). An attribution of this kind should therefore not be drawn from a single training run, and an absolute score reported alone hides which factor is doing the work.

Figure 2 shows PGD-10 robust accuracy across perturbation budgets $\epsilon \in \{2/255, 4/255, 8/255, 16/255\}$ on the full test set.

TABLE 3. Conditional decomposition of white-box robustness on the full test set, mean±std over three runs.

Model	Cond. fooling (%) PGD-10	Cond. fooling (%) AA	Both-correct robust (%) PGD-10	Both-correct robust (%) AA
ResNet-18 AT	**48.58**±0.80	**55.78**±0.23	**59.86**±0.35	**51.89**±0.49
ViT-Tiny AT	55.53±0.50	60.37±0.33	46.11±0.59	41.15±0.39

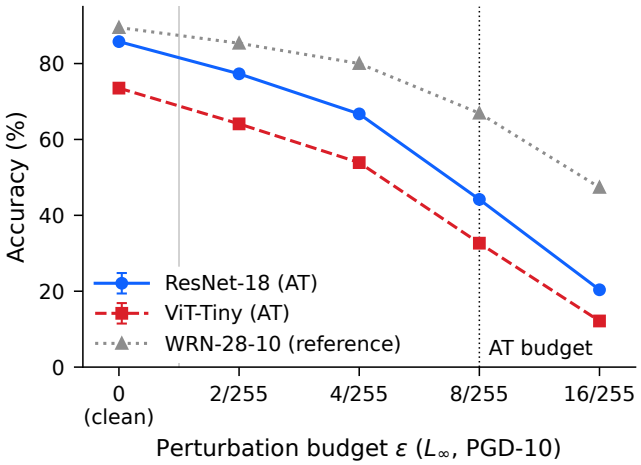


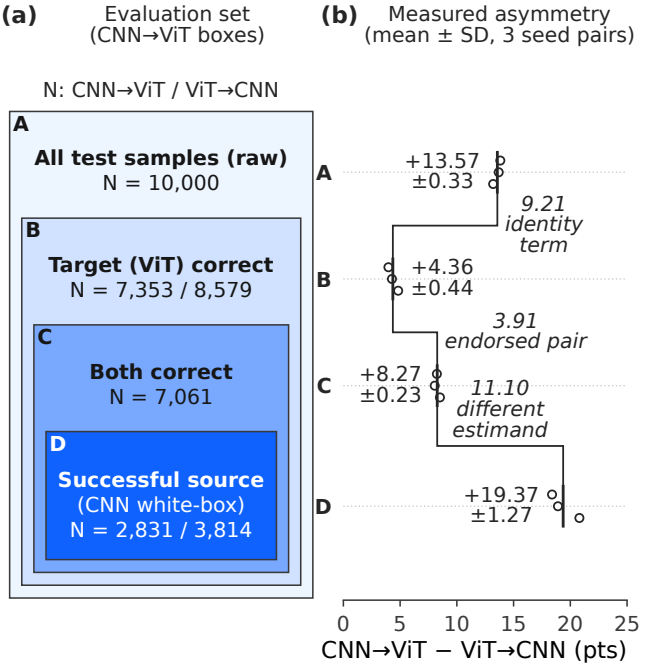
FIGURE 2. PGD-10 robust accuracy against the perturbation budget ϵ on the full test set (mean over three runs; the ± 1 std error bars are narrower than the markers). The horizontal axis is categorical and $\epsilon = 0$ is clean accuracy. The sweep is a separate PGD-10 evaluation, so its $8/255$ point differs from Table 2 by at most 0.08 points. The CNN's margin over the ViT is not constant: it is widest at the small budgets and narrows at $16/255$.

B. FOUR CONDITIONING PROTOCOLS ON FIXED MODELS

We craft PGD-10 adversarial examples on a source model and evaluate them on a target, on the full test set, for each of the

TABLE 4. Transfer asymmetry under the unconditioned rate and three established conditioning protocols (PGD-10, ϵ=8/255, full test set; mean±std over three seed pairs).

Protocol	CNN→ViT	ViT→CNN	Diff. (pp)
Unconditioned (raw)	41.02±0.55	27.45±0.27	+13.57±0.33
Target-correct	19.87±0.18	15.51±0.59	+4.36±0.44
Both-correct	18.25±0.17	9.98±0.31	+8.27±0.23
Successful-source	36.39±0.76	17.02±0.52	+19.37±1.27



Spread 15.01 pts (successful-source minus target-correct)

FIGURE 3. Nested evaluation sets of the four transfer protocols in the CNN→ViT direction (a; boxes are schematic and not area-proportional, *N* is the mean over three seed pairs) and the asymmetry, CNN→ViT minus ViT→CNN, that each protocol yields on the same models and data (b; mean±SD of the paired within-pair difference over three seed pairs, individual pairs as open markers). The ViT→CNN sets nest in the same order (*N* = 10,000, 8,579, 7,061 and 3,814); the unconditioned and both-correct sets are identical in the two directions, the other two are not. The connectors in (b) mark three pairwise distances. On CIFAR-10 the 15.01-point spread is the successful-source excess over both-correct (11.10 points, a different estimand) plus the distance within the endorsed pair (3.91 points); the unconditioned estimate lies inside that range, 9.21 points above target-correct, where Equation (7) places it.

three seed pairs. Because outcomes are logged per sample, the same attack run can be scored under every protocol of Table 1 without recomputation: the protocols differ only in which samples enter the denominator.

The denominators differ sharply between protocols. For CNN→ViT and ViT→CNN respectively they are 10,000/10,000 without conditioning, 7,353/8,579 under target-correct, 7,061 shared samples under both-correct, and 2,831/3,814 under successful-source. On identical models and identical data the asymmetry ranges from $+4.36\pm0.44$ to $+19.37 \pm 1.27$ points, a 4.4-fold ratio between the extremes, and the mean per-pair spread between the largest and the smallest protocol estimate is 15.01 ± 0.84 points. Retraining moves the same quantity far less: across the three training pairs the asymmetry spans 0.63, 0.86, 0.46 and 2.41 points under the four protocols in table order, so the like-for-like comparison, a range against a range, is 15.01 points against at most 2.41. The standard deviation in the Diff. column (0.23 to 1.27 points) is that of the paired within-pair difference, which the two per-direction spreads cannot recover because the directions covary across pairs.

The spread is not a single quantity. On CIFAR-10 it splits into two distances between adjacent protocols (Figure 3b). The successful-source estimate conditions on the source's own white-box failure and therefore estimates a different quantity, how well an attack that already worked carries over; it lies 11.10 ± 1.09 points above the both-correct estimate. The two protocols we endorse for directional claims, target-correct and both-correct, which differ only in whether the source must also be correct, are 3.91 ± 0.32 points apart, with a run-level 95% t interval of $[3.12, 4.70]$ (two degrees of freedom); that distance is the difference between two estimands, not analyst discretion, and the marginal subsets described below account for it. Because target-correct is the lowest and successful-source the highest estimate in every pair, the two distances make up the 15.01-point range. The unconditioned estimate lies inside it, 9.21 ± 0.76 points above target-correct and 5.79 points below successful-source, which is where Equation (7) places it given the targets' clean errors, the conditioned rates and the measured premise p (Section IV-B1). The endorsed distance is 5.97 points on CIFAR-100 (interval $[3.53, 8.40]$), 1.95 points under the L_2 budget ($[1.33, 2.56]$), and 0.82 and 0.49 points in the two SVHN pairs. On CIFAR-100 the unconditioned estimate is the upper end of the range, and the identity term, 13.58 points, makes up nearly all of the 13.83-point spread.

The protocol also changes the statistical reading. Under the both-correct protocol, which scores both directions on the same samples and therefore supports paired inference, the difference is 8.27 ± 0.24 points, differing slightly from the ±0.23 of the table only because the two estimators differ. Each run has its own paired bootstrap CI ($[7.61, 9.44]$, $[7.10, 9.01]$ and $[7.28, 9.20]$). Averaging their endpoints gives $[7.33, 9.22]$, a description of three intervals that carries no coverage guarantee of its own; for inference at the run level, the across-run t interval is $[7.68, 8.86]$. The sign-flip permutation p is below 5×10^{-5} in every seed, and TOST does not establish equivalence at any of the three margins (±1, ±2 and ±3 points), since every interval lies entirely outside ±3.

Protocol names also leave sub-choices open. Under the loose reading of successful-source (Table 1), which also admits samples the source already misclassifies on clean input, the measured asymmetry falls from $+19.37$ to $+14.60$ points and the protocol spread from 15.01 to 10.45 points. That shift of 4.77 ± 0.39 points is larger than the run-to-run standard deviation of any protocol in Table 4 by more than a factor of three, and the other three protocols are unchanged by construction.

What is stable on CIFAR-10 is the direction: CNN-crafted examples transfer better to the ViT than the reverse under all four protocols, in all three seeds, and also under the momentum-based MI-FGSM [41] (19.07 ± 0.33 versus $15.42 \pm 0.52\%$ target-correct; this bidirectional check uses the attack's own default step of $\alpha = \epsilon/10$, not the $\alpha = 2/255$ used elsewhere, which is why its ViT→CNN figure differs slightly from the matched-budget value in Table 14), so the effect is not an artifact of the PGD attack itself. Its magnitude and significance, and with them any equivalence claim, depend on the protocol.

The distance between the two endorsed protocols has a compositional source. Samples that the target solves on clean input but the source does not are three to four times more fragile than jointly solved ones ($41.2 \pm 0.8\%$ and $59.1 \pm 1.7\%$ fooling versus 10.0 and 18.2%), and these marginal subsets differ five-fold in size ($1{,}517 \pm 102$ versus 291 ± 16) because the two models differ in clean accuracy. Conditioning on target-correct admits both marginal subsets, whose asymmetric sizes pull the two rates toward each other; conditioning on both-correct excludes them, and the directional difference reappears. This account covers three of the four protocols. The successful-source protocol selects by the *source's* own robustness, which is not a function of the pair's clean-accuracy gap, so it contributes a second and partly opposing term (Section IV-D4).

1) How much of the raw rate is the target's clean error?

Adding the WideResNet-28-10 reference as both source and target gives a 3×3 matrix for each training pair (Table 5), with six off-diagonal directions in which Equation (7) can be checked against targets of very different clean error. Its premise holds tightly on these adversarially trained targets: across 36 off-diagonal directions (two datasets, three seed pairs, three targets, two sources) the measured p ranges from 0.989 to 1.000. The $p = 1$ form therefore reproduces every measured unconditioned rate to within 0.41 points, with a mean absolute residual of 0.095 points and at most 1.39% of the deviation it explains. That residual is $e\,(1-p)$, the premise restated in points, and not a second confirmation; the informative quantity is p itself. Stated as a correlation between the deviation and the target's clean error ($r = 0.997$), the same result would be understated: a near-unit correlation between a quantity and its own dominant component is arithmetic.

What remains dataset-specific is the proportionality factor. Because $(1 - r_{\text{cond}})$ depends on how transferable the attack is, the fitted slope is 0.762 points per point of target error on CIFAR-10 and 0.656 on CIFAR-100, an average over the targets measured that should be re-estimated whenever the identity is applied to other model pools. Unconditioned transfer rates therefore do not measure transferability across architectures; they measure transferability plus the target's clean weakness, in a proportion that Equation (7) makes explicit.

TABLE 5. 3×3 transfer matrix (PGD-10, ϵ=8/255, full test set; mean±std over three seed pairs; the WideResNet column and row use the single external checkpoint). Raw = unconditioned fooling rate, Cond. = target-correct conditioned rate. Diagonal entries are white-box attacks. The attacks for this matrix were generated in a separate run (attack seed 42); the cells it shares with Tables 3 and 4 differ from them by at most 0.06 points, within the attack-initialization variance of Section IV-H.

Source	Target	Raw (%)	Cond. (%)
ResNet-18 AT	ResNet-18 AT	55.84±0.53	48.52±0.84
ResNet-18 AT	ViT-Tiny AT	41.07±0.58	19.93±0.20
ResNet-18 AT	WRN-28-10	21.69±0.09	12.48±0.10
ViT-Tiny AT	ResNet-18 AT	27.45±0.34	15.52±0.64
ViT-Tiny AT	ViT-Tiny AT	67.34±0.18	55.57±0.44
ViT-Tiny AT	WRN-28-10	18.25±0.12	8.65±0.14
WRN-28-10	ResNet-18 AT	32.63±0.14	21.48±0.31
WRN-28-10	ViT-Tiny AT	39.73±0.25	18.15±0.28
WRN-28-10	WRN-28-10	33.03	25.16

C. SOURCE EFFECT, NOT TARGET WEAKNESS

The 3×3 matrices also say where the direction comes from. A CNN→ViT advantage can arise because CNN-crafted examples transfer better to any target, a source effect, or because the ViT is the easier target for any source, a target effect. We separate the two with an additive model fitted, protocol by protocol, to the six off-diagonal rates of each training pair, $r(s \to t) = \mu + \alpha_s + \beta_t + \varepsilon_{st}$ with sum-to-zero coding, so that the asymmetry equals the source difference $\alpha_{\text{CNN}} - \alpha_{\text{ViT}}$ plus the target difference $\beta_{\text{ViT}} - \beta_{\text{CNN}}$ plus a residual difference. Six cells and five parameters leave one residual degree of freedom, so a high R^2 is expected and carries little evidence: the fit describes each matrix and tests nothing, and the fixed-third-model contrasts below need no model at all.

The source term carries the asymmetry on both datasets (Table 6). On CIFAR-10 the source difference is $+5.13 \pm 0.09$ points and the target difference -2.03 ± 0.51: examples crafted on the ViT transfer worse, and the ViT is the *less* vulnerable of the two targets, so the target term works against the measured asymmetry of $+4.41 \pm 0.45$. CIFAR-100 repeats the pattern at a larger size ($+9.54 \pm 0.64$ against -6.19 ± 0.93). The source term keeps its sign under every protocol we can score, whereas the target term does not. Under both-correct the target difference is -0.53 ± 0.59 on CIFAR-10, indistinguishable from zero, and -2.85 ± 0.93 on CIFAR-100, while the source difference grows to $+6.76 \pm 0.35$ and $+11.79 \pm 0.83$; under successful-source, which the CIFAR-100 records allow, the terms are $+20.82 \pm 1.43$ and -8.12 ± 2.23. The WideResNet supplies the same contrasts without a model. As a fixed target it is fooled by 12.48% of ResNet-crafted and 8.65% of ViT-crafted examples, so ViT-crafted examples transfer worse even to a third architecture; as a fixed source it fools the ResNet at 21.48% and the ViT at 18.15%, so the ViT is also the harder target for a third model's perturbations. The fixed-model contrasts keep these signs under both-correct ($+4.69 \pm 0.38$ and -2.60 ± 0.72 on CIFAR-10), and the target contrast is negative in every conditioned protocol on both datasets. Scored on raw rates, the same decomposition gives a target difference of $+8.13$ points on CIFAR-10 and

$+10.72$ on CIFAR-100: unconditioned rates count the ViT's larger clean error and make it look like the weaker target. A weaker-target reading cannot rest on the incoming-transfer correlation either. Incoming conditioned transfer does track each target's own white-box conditioned fooling rate ($r = 0.986$ over the three targets), but that correlation is carried by the robust WideResNet, which absorbs attacks from both families (10.57% incoming); between the ResNet and the ViT, incoming transfer differs by 0.54 points (18.50 against 19.04%) while the asymmetry is 4.41.

The 8×8 matrices of Mahmood et al. [8] admit the same decomposition on models trained elsewhere, under their both-correct protocol with $m = 1000$ samples per pair and the strongest of FGSM, PGD and MIM in each cell ($\epsilon = 0.031$). Their regime differs from ours in every respect that matters here: the ViT and BiT-M models are pre-trained on ImageNet-21k and fine-tuned without adversarial training, ResNet-56 and ResNet-164 are trained on CIFAR without adversarial training, and their diagonal is an independently trained copy of the same model, not a white-box attack, so we exclude it. The additive model fits their matrices poorly ($R^2 = 0.368$ on CIFAR-10 and 0.451 on CIFAR-100) because transfer within a family is much stronger than across families; a same-family term raises R^2 to 0.785 and 0.843. Under either fit the family-level source difference favors the ViT on CIFAR-10 (-25.01 points additive, -10.78 with the family term, CNN minus ViT), the opposite of ours, and the ViT is the stronger source in all 12 cross-family pairs that exclude the hybrid ViT-R50. On CIFAR-100 their ResNet-56 and ResNet-164 show our direction, CNN→ViT stronger in $6/6$ pairs, but the target term carries it ($+30.15$ additive, $+16.81$ with the family term) while the source term still favors the ViT (-21.90 and -8.55). The direction agrees with ours there; the mechanism does not. Section V-A discusses what this leaves open.

D. ACROSS DATASETS AND THREAT MODELS

Table 8 collects the four protocols in every setting of this paper. The two endorsed protocols agree in sign within every setting, and the sign changes across protocols only on SVHN, where all four estimates lie within three points of zero. The settings are taken in turn below (Sections IV-D1 to IV-D3), followed by the relation between the spread and the clean-accuracy gap over every pair we measured (Section IV-D4).

1) A second dataset: CIFAR-100

A protocol effect that exists only on CIFAR-10 would not support a general reporting recommendation, so we repeated the whole design on CIFAR-100: same architectures, same threat model, same attack budget, same three-seed structure, same four protocols. Three directional predictions and three endpoint bands were registered in an append-only document before any measurement from this dataset existed, so that the outcome could not be reinterpreted after the fact. The registration does not precede the start of training; its commit precedes the first reported number, an epoch-5 pilot gate, by sixty-six minutes.

Absolute robustness is lower than on CIFAR-10 and keeps the CNN-over-ViT ordering (Table 7). Across the same four protocols the measured CNN→ViT asymmetry runs from $+4.96 \pm 1.01$ points (target-correct) to $+18.53 \pm 0.71$ points (unconditioned), with $+10.92 \pm 0.34$ under both-correct and $+17.50 \pm 0.92$ under successful-source. The mean per-seed spread is 13.83 ± 1.30 points, against 15.01 ± 0.84 on CIFAR-10, so the harder dataset leaves the protocol effect in the same range, slightly below the CIFAR-10 value. The direction is positive in all twelve measurements (four protocols $\times$ three seed pairs), as it was on CIFAR-10. Under the both-correct protocol the paired difference is 10.92 points. As on CIFAR-10, the quoted interval $[9.47, 12.35]$ is the average of the three per-seed bootstrap endpoints and has no coverage guarantee; the across-run t interval, which is the one that matches a claim about architectures, is $[10.07, 11.77]$. The sign-flip permutation p is below the resolution of $20{,}000$ permutations in every seed, and TOST does not establish equivalence at any margin tested.

One of the three registered predictions is confirmed, one fails under the reading we treat as binding, and the third is not testable in this design. The prediction that the sign would be preserved is confirmed ($12/12$). The prediction that the protocol spread would be larger on the harder dataset was registered as a comparison against the CIFAR-10 spread, quoted at the time as 10.45 points. Against that literal threshold it holds ($13.83 > 10.45$); against the CIFAR-10 spread as it now stands under the strict successful-source reading it does not ($13.83 < 15.01$). We count this as a failure, since the registration named the CIFAR-10 spread and not the number. The third predicted that the deviation of the unconditioned from the conditioned rate would grow with the target's clean error. The measured slope is $+0.656$ points per point of target error, positive as predicted; but on CIFAR-100 two of the three targets have clean errors that differ by 0.21 points (36.15 and 36.36), so the correlation is computed over two effective clusters, not three, and its value ($r = 0.931$ over six directions, 0.974 aggregated to targets) rests on a two-point contrast. The measurement is *consistent with* the prediction without *testing* it, and it does not count as a third confirmation.

One registered endpoint was missed: we predicted clean accuracy in the 33 to 40% band for the adversarially trained ViT and measured $43.17 \pm 1.10\%$. The band assumed that the clean-accuracy cost of adversarial training scales the same way for both architectures. It does not: the ViT retained 0.885 of its clean pre-training accuracy under adversarial training, against 0.802 for the ResNet. The two adversarial endpoints, which are the ones the protocol claim depends on, fell inside their bands (PGD-10 predicted 10 to 14%, measured 11.15; AutoAttack predicted 8 to 11%, measured 8.87). Conditioning also changes the class mix, so an asymmetry could in principle be a composition artifact; it is not. Under target-correct the composition component of the asymmetry

TABLE 6. Additive source–target decomposition of transfer matrices (points; off-diagonal cells). Source difference = $\alpha_{\text{CNN}} - \alpha_{\text{ViT}}$ (positive: CNN-crafted examples transfer better); target difference = $\beta_{\text{ViT}} - \beta_{\text{CNN}}$ (positive: the ViT is the more vulnerable target); asymmetry = CNN→ViT minus ViT→CNN. Upper block: our 3 × 3 matrices (ResNet-18 AT, ViT-Tiny AT, WideResNet-28-10; in the standard-trained row the clean-trained ResNet-18 and ViT-Tiny of each pair with RobustBench's standard WideResNet-28-10), fitted per training pair and protocol, mean±std over three pairs; the matrices come from the separate attack run of Table 5, so the CIFAR-10 target-correct asymmetry is 4.41 here against 4.36 in Table 4, and with six cells and five parameters R^2 describes and does not test. Middle block: the WideResNet as a fixed third model, with no fit: as a target it gives a source contrast (rate from the CNN minus rate from the ViT), as a source a target contrast (rate to the ViT minus rate to the CNN). Lower block: the 8 × 8 both-correct matrices of Mahmood et al. [8] (standard-trained models, m=1000, diagonal excluded), CNN and ViT families without the hybrid ViT-R50 unless stated, fitted with and without a same-family term.

Data	Protocol or fit	Source diff.	Target diff.	Asymmetry	Max |resid.|	R^2
Our matrices, additive fit per training pair						
CIFAR-10	Target-correct	+5.13±0.09	−2.03±0.51	+4.41±0.45	0.65±0.05	0.9782±0.0032
CIFAR-10	Both-correct	+6.76±0.35	−0.53±0.59	+8.30±0.16	1.04±0.11	0.9643±0.0072
CIFAR-10	Unconditioned	+4.47±0.10	+8.13±0.33	+13.62±0.38	0.51±0.03	0.9964±0.0004
CIFAR-100	Target-correct	+9.54±0.64	−6.19±0.93	+4.96±1.02	0.80±0.05	0.9871±0.0015
CIFAR-100	Both-correct	+11.79±0.83	−2.85±0.93	+10.92±0.34	0.99±0.10	0.9851±0.0035
CIFAR-100	Successful-source	+20.82±1.43	−8.12±2.23	+17.50±0.92	2.40±0.06	0.9830±0.0008
CIFAR-100	Unconditioned	+6.48±0.38	+10.72±0.97	+18.53±0.71	0.67±0.01	0.9946±0.0003
C10, standard-trained	Target-correct	+69.18±2.31	−75.05±1.72	−4.49±0.85	0.69±0.20	0.9996±0.0002
WideResNet-28-10 as a fixed third model, no fit						
CIFAR-10	Target-correct	+3.83±0.17	−3.33±0.59	–	–	–
CIFAR-10	Both-correct	+4.69±0.38	−2.60±0.72	–	–	–
CIFAR-100	Target-correct	+7.94±0.54	−7.79±0.95	–	–	–
CIFAR-100	Both-correct	+9.81±1.03	−4.83±0.75	–	–	–
CIFAR-100	Successful-source	+16.03±1.42	−12.91±2.29	–	–	–
C10, standard-trained	Target-correct	+67.80±2.11	−76.43±2.02	–	–	–
Mahmood et al. [8], 8 × 8 both-correct matrices, family level						
CIFAR-10	Additive	−25.01	+2.08	−24.01	74.04	0.3681
CIFAR-10	Same-family term	−10.78	−12.15	−24.01	29.33	0.7852
CIFAR-100	Additive	−22.77	+14.74	−9.02	66.29	0.4508
CIFAR-100	Same-family term	−9.43	+1.39	−9.02	24.33	0.8425
CIFAR-100	ResNet-56/164 vs. ViT, additive	−21.90	+30.15	+9.15	–	–
CIFAR-100	ResNet-56/164 vs. ViT, same-family	−8.55	+16.81	+9.15	–	–

TABLE 7. CIFAR-100 replication (mean±std over three independent training runs per architecture, full test set, ϵ=8/255). Both models are trained with the recipe of Section III-E. Absolute robustness is lower than on CIFAR-10 for both architectures, and the CNN-over-ViT ordering is unchanged.

Model	Clean (%)	PGD-10 (%)	AutoAttack (%)
ResNet-18 AT	63.86±1.05	19.30±0.32	15.04±0.54
ViT-Tiny AT	43.17±1.10	11.15±0.60	8.87±0.83

is −1.249, −1.330 and −1.065 points across the three seed pairs, *negative*, so rebalancing the class mix would make the asymmetry larger; under successful-source it is small and positive in every seed (+0.527, +0.435 and +0.382 points), and under the unconditioned and both-correct protocols it is exactly zero, as the decomposition requires.

This result carries two qualifications. The two adversarial bands held with margins of 0.22 points (AutoAttack) and 0.48 points (PGD-10) at the nearest seed-level edge, smaller than the checkpoint-selection noise on this same dataset: sweeping early-stopping patience and curve smoothing over the CIFAR-100 trajectories, with the training run held fixed, moves the reported PGD-10 accuracy by 0.18 to 1.83 points (mean 0.86). The same nine-cell sweep on CIFAR-10 gives 0.54 to 2.83 points per validation split, and the eighteen-cell figure of Section IV-F is larger still because it also varies the split, a dimension the CIFAR-100 protocol does not have. Second, the same held-out split serves selection at both stages instead of two independent splits, a choice inherited from the CIFAR-10 protocol to keep the two datasets comparable.

2) A third dataset with a near-matched pair: SVHN

CIFAR-10 and CIFAR-100 both place the two architectures far apart in clean accuracy (about 12 and 21 points). SVHN [62] does not. Adversarially trained under the same recipe, the ResNet reaches 94.76 and 93.77% clean accuracy in the two runs (mean 94.27) and the ViT 92.82 and 92.03% (mean 92.42), gaps of 1.94 and 1.74 points (mean 1.84); the two-run standard deviations, 0.70 and 0.56, carry little information, so we give the run values. Mean robust accuracies are 55.52 and 52.30% under PGD-10. We call this pair near-matched, not matched, since the PGD-10 gap is still about three points. If the identity term follows the clean-accuracy gap, the direction itself should become a matter of protocol here.

It does. Two of the four protocols report a CNN advantage and two a ViT advantage (Table 8). The two protocols we endorse for directional claims, target-correct and both-correct, agree with each other and both favor the ViT; the sign change separates them from the unconditioned and successful-source rates. The unconditioned reversal follows arithmetically from Equation (7): the ViT target's clean error exceeds the CNN's by about two points here, which is enough to carry a −1.00 target-correct difference to +0.39 raw. Under both-correct,

TABLE 8. Transfer asymmetry (CNN→ViT minus ViT→CNN, percentage points) under the four conditioning protocols in every setting of this paper (PGD-10; ϵ=8/255 under L_∞, ϵ=0.5 under L_2; full test sets). Settings with three seed pairs give the mean±std across pairs; SVHN has two pairs, so both values are printed instead of a standard deviation. Spread is the per-pair range over the four protocols, averaged over pairs. The two endorsed protocols, target-correct and both-correct, agree in sign within every setting; the sign changes across protocols only on SVHN, where all four estimates lie within three points of zero.

Protocol	CIFAR-10, L_∞ 3 pairs, n=10,000	CIFAR-100, L_∞ 3 pairs, n=10,000	SVHN, L_∞ 2 pairs, n=26,032	CIFAR-10, L_2 3 pairs, n=10,000
Unconditioned (raw)	+13.57±0.33	+18.53±0.71	+0.33 / +0.45	+13.03±0.81
Target-correct	+4.36±0.44	+4.96±1.01	−1.16 / −0.85	+2.12±0.18
Both-correct	+8.27±0.23	+10.92±0.34	−0.34 / −0.36	+4.07±0.36
Successful-source	+19.37±1.27	+17.50±0.92	+2.98 / +2.41	+12.06±0.23
Spread (max − min)	15.01±0.84	13.83±1.30	4.14 / 3.26	10.92±0.82

which we treat as primary because it scores both directions on identical samples, the difference is -0.35 points, with a sign-flip permutation p of 0.105 and TOST equivalence accepted at all three margins, including the tightest one we test (± 1 point). The protocol decides whether such an equivalence claim is available at all: at the ± 2 margin fixed before measurement no successful-source estimate could pass, since both point estimates ($+2.98$ and $+2.41$) lie outside the margin, while both both-correct estimates (-0.34 and -0.36) lie well inside it. The acceptance needs one qualification. SVHN was run with two seed pairs, so the quoted interval $[-0.76, +0.07]$ averages the endpoints of two per-seed bootstraps and no across-run interval is available; the equivalence rests on the within-run comparison and on the close agreement of the two runs, and two runs cannot bound the run-to-run variation. A study reporting only the successful-source rate would announce a 2.70-point CNN advantage on SVHN; one reporting only the target-correct rate would announce a 1.00-point ViT advantage.

The four-protocol spread on SVHN is 4.14 and 3.26 points in the two pairs (mean 3.70), against 15.01 ± 0.84 on CIFAR-10 and 13.83 ± 1.30 on CIFAR-100, while the clean-accuracy gaps are 1.84, $\approx$ 12 and $\approx$ 21 points; the three-protocol spread, which is the identity term, is 1.49 and 1.30 points. The relation is not monotone across the three datasets: the largest gap does not produce the largest spread. What separates SVHN is the near-matched pair. Once the spread is small enough it no longer clears the asymmetry itself, so the sign becomes a property of the protocol and not of the architectures, and the direction reported as stable in Section IV-B is stable *in the regime where the two models differ in clean accuracy*, which both CIFAR datasets occupy. SVHN was run in a reduced configuration fixed in advance: two seed pairs instead of three, 50 adversarial-training epochs, no AutoAttack, and no third architecture, so the 3×3 control of Section IV-B1 is not built here.

3) A second threat model: L_2

To test whether the protocol effect is an L_∞ artifact, we re-evaluated the same CIFAR-10 checkpoints under an L_2 budget of $\epsilon = 0.5$, registering three directional predictions before looking at any L_2 output. The models are L_∞-trained, so this probes how the protocol affects the conclusion, not L_2 robustness, and the absolute values below are not comparable with L_2-trained references.

TABLE 9. Same checkpoints, L_2 budget (ϵ=0.5; mean±std over three seed pairs). PGD-L2 uses the full test set; AutoAttack-L2 uses a 5,000-sample subset fixed before measurement, so its clean column differs slightly from the full-set value and the two are not drawn from the same samples.

Model	Clean	PGD-L2	Clean (5k)	AA-L2
ResNet-18 AT	85.78	63.78±0.38	85.33	60.61±0.42
ViT-Tiny AT	73.53	51.72±0.23	73.61	49.85±0.20

Under the changed norm the protocol spread shrinks to 10.92 points, against 15.01 under L_∞, yet stays far above both the two-point floor set in the registration and the run-to-run standard deviation, and the sign is positive in all twelve measurements, so the first two registered predictions hold. The four protocols (Table 8) report asymmetries of $+13.03 \pm 0.81$ (unconditioned), $+2.12 \pm 0.18$ (target-correct), $+4.07 \pm 0.36$ (both-correct) and $+12.06 \pm 0.23$ (successful-source). The third prediction, that the deviation of unconditioned from conditioned rates would rise with the target's clean error, cannot fail once Equation (7) holds, and the fitted slope of $+0.914$ estimates $1 - r_{\text{cond}}$ instead of testing anything; we report the slope rather than a correlation, because the eighteen directions take only seven distinct target-error values, six of them the same fixed WideResNet. The prediction was registered before the identity was written down and now serves as an out-of-sample check on it. The slope also sits where the identity says it should: conditioned rates are far lower under this L_2 budget than under L_∞ (roughly 2 to 9% against 9 to 22%), so the identity predicts a steeper slope, and the measured 0.914 exceeds the 0.762 of the same models under L_∞. Absolute robustness (Table 9) keeps the CNN-over-ViT ordering under both attacks, with McNemar $p < 10^{-57}$ in every seed pair.

4) Spread and the clean-accuracy gap

The settings above suggest that the spread grows with the clean-accuracy gap, and the analysis registered before the CIFAR-100 and SVHN measurements tested this on the direction pairs of all final models (20 direction pairs from eight seed-and-dataset clusters, cluster bootstrap). The four-protocol slope, -0.133 with interval $[-0.570, +0.469]$, includes zero, and we withdraw it. The

three-protocol slope, which leaves out successful-source, is $+0.470$ $[+0.361, +0.665]$. That slope is not independent evidence for a mechanism. In 19 of the 20 direction pairs the three-protocol spread equals the absolute difference between the unconditioned and target-correct asymmetries, because the both-correct estimate lies between them, and that difference agrees with the prediction of Equation (7) to within 0.16 points; the identity term alone gives a slope of $+0.471$ $[+0.3615, +0.6666]$. The three-protocol slope is the identity measured a second time, and we read it as an out-of-sample check of Equation (7).

The pool first used for this fit excluded the SVHN pair. That departs from the registration, which placed SVHN in the fitted arm and named it as the pair that would fill the empty band of small gaps; the reassignment to an independent consistency check was written after the SVHN result was in hand, so the SVHN-inclusive fit above is the binding one. On the SVHN-excluded pool the four-protocol slope is -0.528 $[-0.664, -0.418]$ and the three-protocol slope $+0.431$ $[+0.333, +0.633]$. The opposite signs come from the successful-source protocol, which is the extreme of the four in 16 of the 18 pairs there, with target-correct against successful-source the widest pair (23.77 points average gap, widest in 13 of the 18 directions). A controlled arm calibrates the same relation inside training trajectories: one member of a pair is held fixed and the other is swept along its own trajectory, so architecture, seed, dataset and budget stay constant while the gap moves (116 points from eight clusters, cluster bootstrap at the trajectory level). Its three-protocol slope, $+0.673$ $[+0.602, +0.727]$, has the sign of the observational one and carries the same caveat, since it too is dominated by the identity term. Its four-protocol slope, $+0.273$ $[+0.219, +0.371]$, lies entirely inside the observational interval on the binding pool, so on that quantity the arms neither agree nor disagree, and restricting the observational pool to the matched architecture pair moves the four-protocol slope to -0.100 $[-0.464, +0.092]$. A quantity this sensitive to which pairs enter the fit is not one a mechanism claim can rest on. With six to eight clusters a percentile cluster bootstrap undercovers, so all of these intervals understate their width, and the WideResNet, a single fixed checkpoint, appears in every observational cluster.

Figure 4 extends the relation to every pair we measured: the adversarially trained pairs of the three datasets and both threat models, their pairings with the WideResNet, the same CIFAR-10 pairs at their last checkpoint (Section IV-F), the standard-trained pairs and the external pool of Section IV-E, 60 unordered pairs in all. The CIFAR-100 and L_2 pairings with the WideResNet are not included, and the three pool pairs that coincide with the first CIFAR-10 training pair are counted once, in the adversarially trained groups. The three-protocol spread rises with the gap (slope $+0.796$, Pearson $r = 0.957$; between $+0.767$ and $+0.855$ when any one group is left out), and it equals the unconditioned-minus-target-correct difference at 57 of the 60 points. The prediction of Equation (7), drawn as open rings, tracks every point to

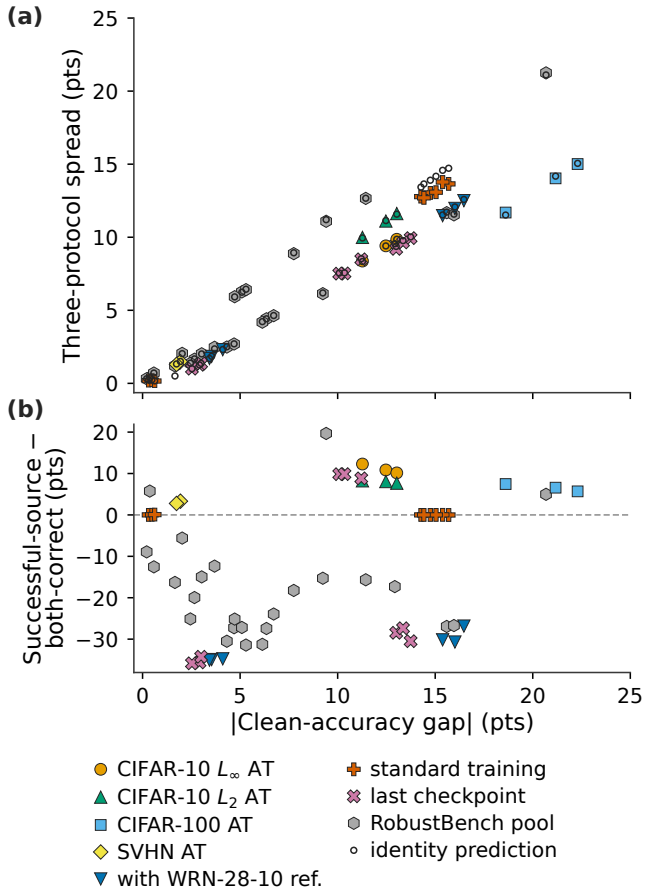


FIGURE 4. Protocol spread against the absolute clean-accuracy gap for all 60 measured model pairs. (a) Spread of the asymmetry over the unconditioned, target-correct and both-correct protocols; small open rings mark the prediction of Equation (7) with $p = 1$ for the same pair. (b) Successful-source minus both-correct asymmetry for the same pairs, the term that the three-protocol spread leaves out. Markers separate the adversarially trained pairs of each setting, their pairings with the WideResNet-28-10 reference, the same pairs at their last checkpoint, the standard-trained pairs and the RobustBench pool; pairs share models, so no fitted line is drawn.

within 1.09 points (mean absolute deviation 0.13), and the largest departures fall on standard-trained targets, where p is furthest from one (Section IV-E). The four-protocol spread does not follow the gap (slope $+0.092$, $r = 0.053$), and among robust pairs of nearly equal clean accuracy it stays large: 9.03 points for the two WideResNet-28-10 checkpoints of Carmon et al. [63] and Gowal et al. [54] (0.21 points apart), 12.73 for the XCiT-S12 against the Gowal et al. model (0.58) and 17.50 for our ResNet-18 against the Engstrom et al. [64] ResNet-50 (1.66). Panel (b) shows why: the successful-source term is unrelated to the gap, close to zero for the standard-trained pairs, whose PGD-10 robust accuracy is near zero so that almost every source attack succeeds, and large and of either sign among robust models. These slopes describe the pool and test nothing, because the pairs share models and runs.

The matched-pair prediction is a consequence of the identity, not a separate finding. The unconditioned, target-correct

and both-correct estimates differ through terms that scale with the clean errors, so a pair matched in clean accuracy should leave them close together whatever its architectures. What the data contribute is the three measured facts the prediction needs: p is at least 0.984 on every adversarially trained target we examined, the both-correct estimate lies between the other two at 57 of the 60 points, and the successful-source term does not depend on the gap. Accordingly SVHN, at a 1.84-point gap, has a three-protocol spread of 1.49 and 1.30 points in its two pairs, and the standard-trained ResNet-18 and WideResNet-28-10, 0.47 ± 0.16 points apart, have all four protocols within 0.19 ± 0.02 points (Section IV-E). Of the 15-point CIFAR-10 range, the parts that come from the clean-accuracy difference, the identity term and the distance within the endorsed pair, should shrink for pairs matched in clean accuracy; the successful-source excess need not, as the robust pairs above show.

E. STANDARD-TRAINED TARGETS AND AN EXTERNAL MODEL POOL

Two questions remain that our own adversarially trained pair cannot answer: whether the premise p holds on standard-trained targets, the regime most transfer studies attack, and whether the matched-pair prediction holds for models trained by others. Both are answered without new training. First, the clean-trained ResNet-18 and ViT-Tiny checkpoints of the three training pairs, the starting points of adversarial fine-tuning, are combined with the standard-trained WideResNet-28-10 distributed by RobustBench [14] (identifier `Standard`) into one 3×3 matrix per pair. Second, an eight-model CIFAR-10 pool joins our adversarially trained ResNet-18 and ViT-Tiny of the first pair, the standard-trained WideResNet and five robust RobustBench checkpoints, listed here by their RobustBench identifiers: a PreAct ResNet-18 (`Wong2020Fast`) [65], a ResNet-50 (`Engstrom2019Robustness`) [64], a WideResNet-28-10 trained with unlabeled data (`Carmon2019Unlabeled`) [63], an XCiT-S12, which combines attention with convolutional components (`Debenedetti2022Light_XCiT-S12`) [56], and the WideResNet-28-10 reference of the previous sections [54]. Every direction is attacked with PGD-10 at $\epsilon = 8/255$ on the full test set with per-sample logging, and all four protocols, p and the $p = 1$ residual are computed from those records (Table 10).

On standard-trained targets the premise weakens. Over the 18 off-diagonal directions of the three clean matrices, p ranges from 0.877 to 1.000; it is lowest when the standard-trained WideResNet is the target (0.877 to 0.996) and lies between 0.923 and 0.934 for the clean ViT-Tiny target. The $p = 1$ form then misses the unconditioned rate by up to 1.58 points, on the ViT-Tiny target, whose clean error is the largest of the three. There Equation (7) is an upper bound, and the general form, Equation (6), with measured p, is the correction to use. In the pool, where seven of the eight targets are adversarially trained, $p \geq 0.984$ on every adversarially trained target and $p \geq 0.933$ on the standard-trained one, and the $p = 1$ residual stays within 0.35 points over all 56 directions; at the last checkpoint of our own pairs it stays within 0.20 (Section IV-F).

The standard-trained matrices also test the matched-pair prediction under a different training regime. The clean ResNet-18 and the standard WideResNet differ by 0.47 ± 0.16 points in clean accuracy, and their four protocol estimates agree to within 0.19 ± 0.02 points (-3.35, -3.54, -3.51 and -3.50 in the first pair). The two unmatched pairs, about 15 points apart (15.16 ± 0.48 and 14.69 ± 0.60), spread by 13.25 ± 0.34 and 13.08 ± 0.60 points, and in every run the unconditioned protocol reverses the sign that the other three report: for the clean ResNet-18 and ViT-Tiny it gives $+7.81$ to $+10.09$ points where target-correct gives -5.22 to -3.56. Under standard training, then, our own ResNet-18 and ViT-Tiny show the opposite direction under every conditioned protocol (both-correct -3.02 to -1.30), the direction Mahmood et al. report, but not their mechanism. The additive fit (Table 6) keeps the clean ViT-Tiny by far the weakest source (source difference $+69.18 \pm 2.31$) and turns the sign through the target term (-75.05 ± 1.72): the clean ViT-Tiny is an even harder target than it is a weak source. The standard WideResNet shows both contrasts directly, being fooled by 81.40% of ResNet-crafted and 13.60% of ViT-crafted examples and fooling the ResNet at 84.05% and the ViT at 7.62%.

In the pool, whose 28 pairs span clean gaps from 0.21 to 20.68 points, the three-protocol spread rises by 0.872 points per point of gap (model leave-out bootstrap interval $[+0.534, +1.274]$, $r = 0.944$), whereas the four-protocol spread shows no reliable relation ($+0.559$, $[-0.205, +4.582]$). The sign changes across protocols in one pair of 28, our adversarially trained ViT-Tiny against the standard-trained WideResNet, which is also the pair with the widest gap (20.68 points; unconditioned -6.72, target-correct $+14.53$). Robust models of nearly equal clean accuracy confirm the matched-pair prediction and its limit at once: for the two robust WideResNet-28-10 checkpoints of Carmon et al. and Gowal et al., 0.21 points apart, the three-protocol spread is 0.32 points and the four-protocol spread 9.03. An additive fit to the pool's target-correct matrix ($R^2 = 0.8955$ over 56 cells) places our ViT-Tiny as the weakest adversarially trained source (source effect -4.87, against $+2.94$ to $+3.65$ for the other robust models), with a target effect of -0.63 inside the range of the robust models (-4.26 to $+2.76$). The robust XCiT-S12, an attention model with convolutional components, is as strong a source as the CNNs ($+3.06$). The source effect of Section IV-C is therefore better read as a property of this ViT-Tiny, trained from scratch to a low clean accuracy, than of transformers as a family; one robust attention-based model in the pool is too few to say more.

F. CHECKPOINT SELECTION AND THE ASYMMETRY

The offline rule that picks the final checkpoint from a fixed trajectory is usually left implicit, although robust overfitting

TABLE 10. Premise p of Equation (6) and the matched-pair prediction outside our adversarially trained pair (CIFAR-10, PGD-10, ϵ=8/255, full test set). Top: p by target type over the off-diagonal directions, with the largest absolute error of the $p = 1$ form in points ("–": not stored per target type). Middle: standard-trained pairs, each the clean ResNet-18 or ViT-Tiny of one training pair or RobustBench's standard-trained WideResNet-28-10, mean±std over three pairs; a sign flip means the four protocols disagree in sign. Bottom: slope of the protocol spread on the absolute clean-accuracy gap over the 28 pairs of the eight-model pool, with a model leave-out bootstrap interval, since pairs share models.

Target	Directions	p, min–max	Max |resid.|
Our AT targets, 3 × 3	36	0.9890–1.0000	0.41
Our AT targets, last ckpt.	18	0.9915–1.0000	0.20
Clean ViT-Tiny	6	0.9231–0.9338	1.58
Clean ResNet-18	6	0.8961–1.0000	0.51
Standard WRN-28-10	6	0.8774–0.9962	0.64
Pool, AT targets	49	0.9846–1.0000	–
Pool, standard WRN	7	0.9330–0.9962	–
Pool, all targets	56	0.9330–1.0000	0.35
Standard-trained pair	**|Clean gap|**	**Spread**	**Sign flips**
ResNet-18 / WRN	0.47±0.16	0.19±0.02	0/3
ResNet-18 / ViT-Tiny	15.16±0.48	13.25±0.34	3/3
ViT-Tiny / WRN	14.69±0.60	13.08±0.60	3/3
Pool spread	**Slope**	**95% interval**	r
Three protocols	+0.872	[+0.534, +1.274]	0.944
Four protocols	+0.559	[−0.205, +4.582]	0.350
Sign flips (four prot.)		1/28 pairs	

makes that choice consequential in adversarial training [58]. Holding a training trajectory fixed and varying only how its final checkpoint is chosen (validation split, early-stopping patience, and whether the validation curve is smoothed before the argmax; eighteen combinations) moves the reported PGD-10 accuracy by 2.62 to 2.85 points per seed for the ResNet and 1.58 to 2.09 points for the ViT. The weights on disk are identical across these cells; only the rule that picks one of them differs. About half of that dispersion comes from one of the three dimensions. Smoothing the validation curve before the argmax lowers test PGD-10 accuracy monotonically in every ResNet run and in two of the three ViT runs, so it is a systematically worse rule; restricted to the split and patience choices that studies actually vary, the dispersion falls to 1.09 times the seed-level standard deviation for the ResNet and 1.42 times for the ViT. The dispersion is quoted in absolute units and deliberately not as a ratio to seed-level dispersion, because that ratio depends on which of the eighteen cells is the reference: across all eighteen it runs from 0.67 to 3.38 for the ResNet and 0.63 to 7.38 for the ViT, inverting the comparison for some of them.

The CIFAR-100 runs supply a counterweight: the epoch selected by our own protocol varies by 25 epochs across the three ResNet seeds, yet the resulting test PGD-10 standard deviation is 0.32 points. The selection *path* was unstable while the selected *outcome* was not; which of the two happens depends on the plateau of the trajectory, not on the architecture.

The selection rule also reaches the transfer asymmetry, because it sets the clean-accuracy gap on which the spread depends. Re-scoring the three training pairs at their last checkpoint, the epoch at which early stopping ended training (epochs 71, 76 and 77 for the ResNets and 45, 38 and 39 for the ViTs, twenty epochs after the selected 51, 56, 57 and 25, 18, 19), instead of the selected one narrows the clean gap from 12.26 ± 0.90 to 10.55 ± 0.59 points (change -1.71 ± 0.84) and the four-protocol spread from 15.01 ± 0.84 to 12.84 ± 0.34 points (change -2.16 ± 0.93); the three-protocol spread falls from 9.21 ± 0.76 to 7.84 ± 0.56. The target-correct asymmetry barely moves ($+4.64 \pm 0.56$ against $+4.36 \pm 0.44$), the successful-source estimate falls from $+19.37$ to $+17.48$, all four protocols keep the CNN→ViT sign in every pair, and the premise holds as before ($p \geq 0.9915$, $p = 1$ residual at most 0.20 points).

The stored training trajectories give the same picture across training. They come from CIFAR-10 retraining runs with the same seeds, not from the selected checkpoints of the main pairs, and from the CIFAR-100 runs themselves, saved about every ten epochs. On CIFAR-10 these retraining runs keep the clean gap at or above 10.59 points at both the selected and the final epoch, so selection moves the gap by about 1.7 points but does not remove it. Its exact size is selection-sensitive, though. Moving one member of a pair from its selected to its final epoch changes the four-protocol spread by -1.53 ± 1.15 points when the ResNet is moved and by -0.03 ± 1.15 when the ViT is, and moving the ViT shifts the target-correct asymmetry by $+2.53\pm0.15$ points; on CIFAR-100 the spread changes by -1.84 ± 0.94 and $+0.84 \pm 0.24$ points. The selection rule thus moves the level of the asymmetry by up to about two and a half points and its spread by about two, without changing the direction or the order of magnitude. It has to be stated in the methods section.

G. THREE READINGS THAT DO NOT SURVIVE DIRECT MEASUREMENT

Architecture comparisons invite three readings beyond the transfer asymmetry: that CNN gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that ViT-specific transfer attacks improve ViT-sourced transfer. Each was measured directly, none holds in this regime, and the same measurements leave two behavioral differences standing.

1) Gradient sparsity without spatial locality

CNN input gradients are sparser than ViT gradients on all three scale-invariant measures (Table 11; per-sample loss gradients so that norms do not depend on the evaluation batch size, alignment as the mean absolute cosine over all $500 \times 499/2$ sample pairs), and the ordering reproduces in every run with seed-level standard deviations at least three times smaller than the gap. This is the one place where our two units of inference disagree, so we report both. Within each run, where the same 500 samples are measured under both models, per-sample paired Wilcoxon tests reject the null on all three measures (Holm-corrected $p < 10^{-11}$, paired Cohen's d between 0.32 and 1.23); at the run level, the unit

TABLE 11. Gradient characteristics computed from per-sample loss gradients (n=500 test samples per run; mean±std over three runs). Bold marks the larger of the two adversarially trained models; the native control is a single checkpoint, so no standard deviation applies.

Metric	ResNet-18 AT	ViT-Tiny AT	ViT (native)‡
Hoyer sparsity	**0.4928**±0.0120	0.4561±0.0055	0.4046
Gini coefficient	**0.6498**±0.0089	0.6099±0.0051	0.5712
Fraction <1% of max	**0.3477**±0.0210	0.2682±0.0090	0.1785
Alignment (absolute)	0.0378±0.0009	**0.0562**±0.0004	0.0794
Alignment (signed)	0.0010±0.0003	0.0004±0.0001	–

‡CIFAR-native ViT control (4×4 patches, 5.4M params); a single checkpoint predating the fixed-split protocol, selected on the test set, included only for the sign of the upsampling effect.

TABLE 12. Spatial locality of input gradients (n=500 samples per run; mean±std over three runs). Paired differences are computed on the same samples. Inference is at the run level: p is a one-sample t-test on the three per-run mean differences, Holm-corrected across the four measures, so that the unit of inference matches the unit of the claim. Lower area and entropy mean more localized; higher Moran's I means more clustered.

Measure	ResNet-18	ViT-Tiny	Diff. (R−V)	p (Holm)
Area for 50% energy	0.0378±0.0017	0.0397±0.0010	−0.0019±0.0027	0.54
Area for 90% energy	0.2345±0.0078	0.2570±0.0067	−0.0224±0.0133	0.40
Spatial entropy (nat)	5.2014±0.0387	5.2524±0.0197	−0.0509±0.0580	0.54
Moran's I	0.4141±0.0132	0.3941±0.0024	+0.0200±0.0156	0.47

an architectural claim needs, three paired runs give raw p of 0.036, 0.042 and 0.067 for Gini, the sub-1% fraction and Hoyer, which Holm correction raises to 0.108. The ordering is consistent in every run and not significant at the run level, so we report it as reproducible and not as significant. It is a product of adversarial training: in the clean-trained counterparts of the same runs the *ViT* is the sparser model (Hoyer 0.380 ± 0.009 versus 0.347 ± 0.002), in line with the sparser feature attributions that Chalasani et al. [29] report for adversarially trained models, and the CIFAR-native ViT control is less sparse than the upsampled ViT on every measure, so the ordering is not an artifact of the $32 \rightarrow 224$ upsampling. Cross-sample alignment is higher for the ViT in absolute cosine, a difference that does reach run-level significance ($p = 0.0009$) and that the clean-trained pair already shows (0.058 versus 0.029); the *signed* mean cosine is near zero for both models ($\leq$ 0.0014 in every run), so samples share sensitivity axes without sharing a consistent direction.

Sparsity counts how many gradient components are large; it does not say whether they lie together in the image, which is what a description of gradients as localized asserts. On the channel-summed energy map of the same gradients, none of four scale-invariant descriptors differs at the run level (Table 12; Holm-corrected p between 0.40 and 0.54). The area holding 90% of the energy even points toward a *smaller* area for the CNN by two percentage points of image area, and two of the four spatial orderings change sign across runs, whereas the sparsity ordering holds in every run on every measure. Adversarially trained CNN gradients place more of their mass on fewer pixels, but those pixels are not more spatially grouped than the ViT's, and we restrict our own

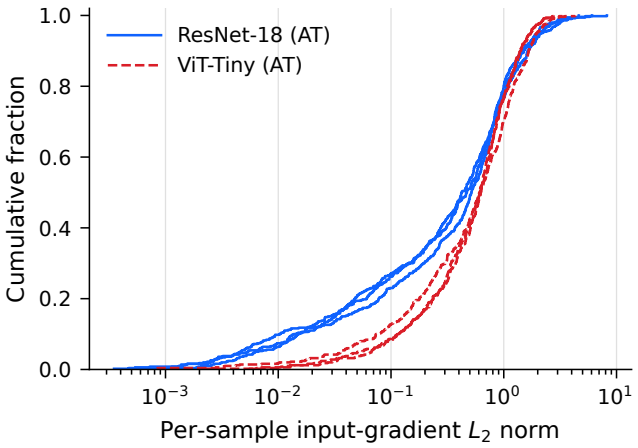


FIGURE 5. Empirical distribution of per-sample input-gradient L_2 norms (per-sample cross-entropy loss, the first 500 test images, one curve per training run, three runs per architecture; logarithmic axis). Magnitude maps for four of these inputs are shown in Figure 6.

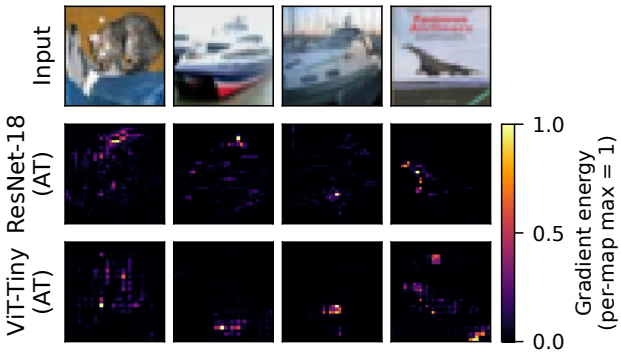


FIGURE 6. Input-gradient magnitude maps for the same test images under both models (seed pair 1; the first four test images; each map is scaled to its own maximum, so magnitudes are not comparable across panels). CNN gradients place more of their mass on fewer components (Table 11); Table 12 shows that this is not accompanied by a difference in spatial arrangement, so the panels should be read as sparsity and not as localization.

claims to sparsity. Figure 5 adds the distribution of per-sample gradient norms, which the scale-invariant measures do not capture, and the magnitude maps of Figure 6 should be read as sparsity, not as localization.

2) Attention entropy and feature drift

Where adversarial damage accumulates along depth differs between the architectures (Table 13; residual block outputs, $n = 1000$ per run), extending layer-wise analyses of clean-trained ViTs [30] to the adversarially trained setting. In the ViT the drift is gradual and direction-only: similarity declines to a minimum at block 11 and recovers slightly at the output, while feature norms stay within 0.45% of their clean values. In the CNN the damage is concentrated: the first block of the last stage falls to 0.8777 ± 0.0181 with a $-13.14 \pm 1.89\%$ norm contraction, and the final block partly repairs the representation (0.9108 ± 0.0060). The measured ViT profile depends on token aggregation. At their minima

TABLE 13. Block-wise clean-to-adversarial cosine similarity under each model's own white-box PGD-10 attack (n=1000 per run; mean over three runs). All summaries are taken at the residual block output, so the ViT and CNN columns are measured at equivalent points; the ViT columns differ only in how a block's tokens are aggregated. Figures for the MLP sub-block output are given in the text. Run-to-run standard deviations are at most 0.0061 in the ViT columns and 0.0181 in the ResNet column (stage 4, block 1); bold marks the minimum of each column.

	ViT-Tiny AT			ResNet-18 AT	
Block	CLS	Patch mean	All tokens	Stage.block	Cosine
1	0.9993	0.9980	0.9927	1.1	0.9951
2	0.9950	0.9959	0.9901	1.2	0.9918
3	0.9902	0.9940	0.9881	2.1	0.9879
4	0.9873	0.9919	0.9861	2.2	0.9853
5	0.9845	0.9904	0.9847	3.1	0.9753
6	0.9814	0.9894	0.9835	3.2	0.9607
7	0.9782	0.9885	0.9826	4.1	**0.8777**
8	0.9743	0.9875	0.9816	4.2	0.9108
9	0.9703	**0.9870**	0.9809		
10	0.9661	0.9872	0.9805		
11	**0.9604**	0.9874	**0.9803**		
12	0.9663	0.9915	0.9837		

the CLS pathway drifts about three times as far as the patch-token mean (0.9604 versus 0.9870, a ratio of 3.05 in $1-\cos$), at different depths (blocks 11 and 9), and read at the MLP sub-block output instead of the residual block output the minima move to 0.9343 at block 10 and 0.9840 at block 8. Layer-wise drift curves from different papers are comparable only when both the aggregation and the tensor are stated.

The attention reading does not hold. Adversarial perturbation leaves the entropy of the CLS attention distribution unchanged: the largest per-layer mean change is 0.0045 nats, against clean layer means between 4.61 and 5.11 nats, and with a margin of ± 0.05 nats fixed from the scale of the measure, run-level TOST over the three runs accepts equivalence in all 12 layers, the weakest at $p = 0.006$ (Figure 8a). What grows with depth is the displacement of the attention map, from 0.028 total variation in the first layer to 0.084 in the last (panel b), so the attack redirects attention slightly without broadening or sharpening it; the single-sample maps of Figure 7 show redirection, not diffusion, and an earlier 8-sample figure batch had suggested an attention effect that this measurement does not support. Feature drift does not separate flipped from unflipped samples either; the unflipped group here also contains the samples a model already misclassifies on clean input. The last-block cosine is 0.9895 for flipped against 0.9877 for unflipped ViT samples, and 0.9058 against 0.9143 for the CNN; neither difference is significant at the run level (paired t over three runs, $p = 0.20$ and $p = 0.080$), although the CNN difference has the same sign in all three runs. Attention-shift effects under patch perturbations [33] and attacks on the attention mechanism itself [32] belong to a different threat model.

3) A ViT-specific transfer attack

A natural objection to the transfer results is that gradient attacks treating the ViT as a generic differentiable model

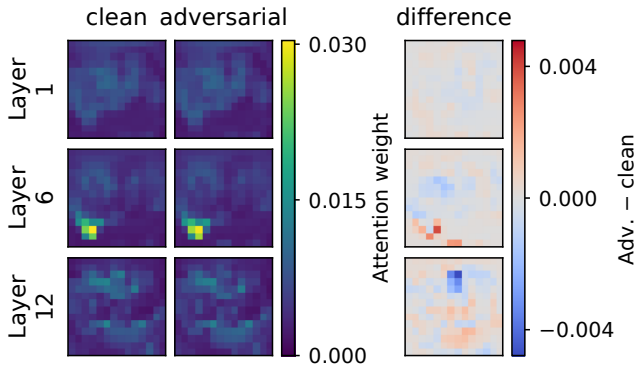


FIGURE 7. CLS-token attention maps (post-softmax, head-averaged) across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12); columns show clean attention, adversarial attention and their difference, with one shared color scale for the attention columns and one for the difference column. A single sample is shown for illustration (the first test image, seed pair 1); the depth trend in attention displacement is measured over n=1,000 samples per run in Figure 8(b).

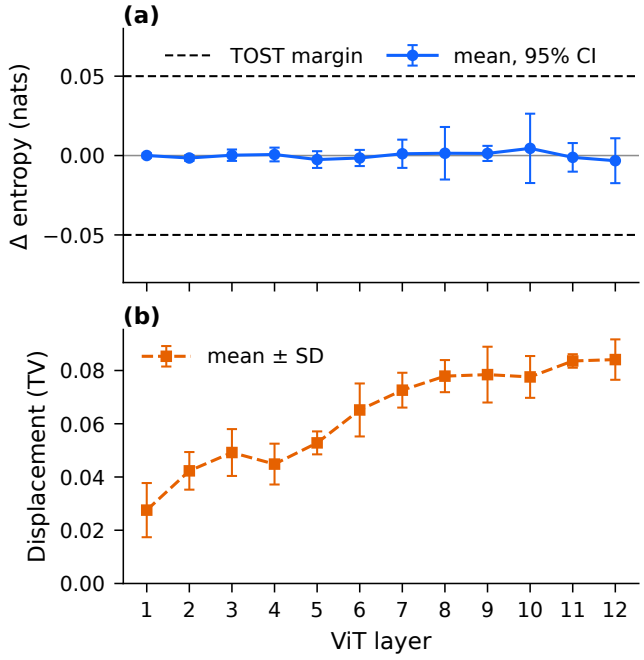


FIGURE 8. (a) Change in CLS attention entropy under attack (adversarial minus clean) per layer, mean over three runs with 95% t confidence intervals; the dashed lines mark the ± 0.05-nat TOST margin, and every interval lies inside it. (b) CLS attention displacement (total variation between the clean and adversarial maps) per layer, mean$\pm$std over runs. n=1,000 samples per run; single-sample maps are shown in Figure 7.

TABLE 14. ViT$\rightarrow$CNN transfer under Token Gradient Regularization (TGR) [45] and plain momentum iterative FGSM (MI-FGSM) at a matched budget (L_∞, ϵ=8/255, 10 steps, α=2/255; n=10,000 per run; mean$\pm$std over three runs).

Measure (%)	TGR	MI-FGSM
White-box on source (raw)	51.56±0.80	**66.43**±0.29
Transfer, target-correct	12.70±0.41	**15.71**±0.60
Transfer, both-correct	7.93±0.42	**10.08**±0.30

handicap the ViT-sourced direction, since attacks designed for transformer structure are reported to transfer better. Token Gradient Regularization (TGR) [45], which suppresses extreme-valued token gradients in the attention, QKV and MLP paths of the source during backpropagation, tests the objection; our adaptation applies it to the CIFAR-10 ViT-Tiny at the budget used throughout (Section III-C), with plain MI-FGSM generated in the same run as a paired control. TGR does not improve transfer in this regime; it degrades it (Table 14). It gives up 14.9 points of white-box strength on the source, and conditioned transfer falls by 3.0 points under target-correct and by 2.2 under both-correct, with paired McNemar tests on the both-correct subset favoring MI-FGSM in every run ($p = 6.1 \times 10^{-10}$, 1.1×10^{-27}, 7.0×10^{-34}). The targets are adversarially trained, the inputs are 32×32 CIFAR-10 images upsampled inside the model and the source is a 5.7M-parameter ViT-Tiny, whereas TGR was built for standard-trained ImageNet ViTs, so the result bounds our adaptation in this regime and not the method in general. For this paper its consequence is that the CNN→ViT asymmetry is not an artifact of architecture-agnostic attacks: giving the ViT direction a transformer-specific attack makes it weaker.

H. SOURCES OF VARIANCE

Four sources of variance bear on the comparisons above. Attack-initialization stochasticity is negligible: re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed 1,000-sample subset leaves clean accuracy exactly unchanged, since the loader does not shuffle, and moves robust accuracy by a standard deviation of 0.05 points for the CNN and 0.00 for the ViT. Training-run variance is larger but still small: across the three independent runs per architecture, the standard deviation for the adversarially trained pair is at most 0.55 points on clean accuracy and 0.50 on PGD-10 (Table 2).

Comparing the protocol effect against those two figures directly would be an error of scale, because the protocol effect is measured on the transfer asymmetry while both figures above are measured on absolute accuracy. We therefore quantify the protocol and training-run effects on the *same* quantity, the off-diagonal asymmetry, and report both in absolute units. Within a fixed seed pair, changing only the conditioning protocol moves the asymmetry by 15.01 ± 0.84 points. This figure is the mean of the per-seed ranges; it coincides with the range of the protocol means here because the extremal protocols are the same in all three seeds, and differs on CIFAR-100 (13.83 against 13.58) because there they are not. Retraining both architectures moves that same asymmetry across a range of at most 2.41 points, with a standard deviation of 0.23 to 1.27 points depending on the protocol it is measured under (Section IV-B). Fifteen points against at most two and a half is the ordering the paper rests on. Changing the checkpoint-selection rule moves the four-protocol spread by 2.16 points on average (Section IV-F), about as much as retraining moves the asymmetry within a protocol (2.41 at most), so it leaves that ordering intact.

We do not headline the ratio of these two spreads. With three runs the denominator carries two degrees of freedom, so a χ^2 interval on any single ratio is very wide: for the successful-source protocol it runs from 0.8 to 9.9 and includes unity. Reported as a sensitivity, the ratio spans 5.2 to 32.6 across dispersion measures and reference protocols, and equals 28.3 under the like-for-like comparison on the both-correct protocol we treat as primary. Two cautions bound it further: the protocol with the largest asymmetry also carries the largest run-to-run standard deviation, so numerator and denominator are not independent, and because all three runs share one validation split the denominator excludes split-to-split variance, which makes the ratio an upper bound. The absolute comparison survives both, which is why it and not the fold-change carries the claim. The fourth source, the checkpoint-selection rule, is treated in Section IV-F.

The three CIFAR-10 training pairs and their CIFAR-10 evaluations, like the earlier checkpoint pair mentioned in Section III-E, were run on an NVIDIA RTX 5060 Ti (16GB VRAM); the CIFAR-100, SVHN and L_2 campaigns, the checkpoint-selection trajectories and the standard-trained, last-checkpoint and RobustBench-pool transfer runs were run on an RTX 5090 (32GB). All runs used PyTorch 2.6.0 with CUDA 12.8 and deterministic cuDNN settings. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on the full 10,000-sample test set in seeded 1,000-sample chunks with per-sample outcome logging. The three-seed training pipeline required 21.7 GPU-hours in total.

V. DISCUSSION

A. WHAT THE MEASUREMENTS SHOW

The central result is an ordering of dispersions on a single quantity. Changing only the conditioning set moves the measured transfer asymmetry by 15.01 ± 0.84 points on fixed models, whereas retraining both architectures moves it across a range of at most 2.41 points, with standard deviations of 0.23 to 1.27 (Section IV-H). For scale on a different quantity, re-running an attack with a new random initialization moves absolute robust accuracy by about 0.05 points and retraining moves the adversarially trained pair by at most 0.55. A number that the measurement choice moves this far, reported without its protocol, tells a reader little about the architectures.

One objection deserves an answer in its own terms: the four protocols estimate four different quantities, so the distance between them is not a variance, and calling it one would be a category error. We agree, and we make no assumption that the architectures have a single true asymmetry that the protocols estimate with error. Our claim concerns what a reader can recover. When a study reports one number without naming the protocol, the reader cannot tell which of the four quantities was estimated, and the spread we measure is the size of that ambiguity, a bound on what the omission costs. Conditioning itself remains established practice [11], [40], [41] and is necessary; the point is that the choice of conditioning set has to be reported with the result, in the main text.

Whether the direction generalizes is a separate question, and the comparison with Mahmood et al. [8] answers it only in part. They report the opposite direction, transfer out of the transformer stronger than transfer into it, and Benz et al. [7] and Shao et al. [9] report the same ordering on their own model sets. The protocol does not explain the disagreement on CIFAR-10. Rescored with Equation (7), their BiT-M-R50x1 and ViT-L-16 pair would move by -0.59 points between both-correct and unconditioned scoring (range given $p = 1$, allowing for the unknown share of samples the source misclassifies: $[-1.25, +2.15]$) against a both-correct asymmetry of -36.80, and none of their 12 cross-family CIFAR-10 pairs changes sign. The two targets of that pair misclassify at most 2.5% of clean inputs (clean accuracies 97.5 and 99.1%), so the result barely depends on p: at the lowest p we measured on adversarially trained targets the effect is -0.57. On CIFAR-100 the rescoring is less conclusive. Its central estimate predicts a sign reversal under raw scoring in 6 of the 12 cross-family pairs, all of them ResNet-56 or ResNet-164 against a ViT, but every one of those ranges includes zero, and the substitution of both-correct for target-correct rates is weakest there, since these sources misclassify up to 28.40% of clean inputs. That rescoring substitutes the both-correct rate for the target-correct one, an approximation that errs by 3.54 points on our own pair, whose source clean errors are larger than any of theirs. The source-target decomposition of Section IV-C gives a partial reconciliation instead. On CIFAR-100 their scratch-trained ResNet-56 and ResNet-164 show our direction against all three non-hybrid ViTs, carried by a target term, while their pre-trained BiT-M models show the opposite direction, which the additive fit assigns to the source term; and our own clean-trained ResNet-18 and ViT-Tiny reach Mahmood et al.'s direction by the same route, through the target term, while the ViT-Tiny stays by far the weaker source (Section IV-E). Demontis et al. [39] trace transferability to the target's vulnerability and the surrogate's complexity; in our adversarially trained matrices it is the source side that decides the direction. Which difference between the regimes reverses the source effect, ImageNet-21k pre-training or model scale, remains open. A shared protocol with released per-sample records would let it be settled directly.

The four protocols are not interchangeable, and each has a defensible use. The unconditioned rate should not be used for architecture comparison at all: its deviation from the target-correct rate is the target's clean error entering through the law of total probability, exactly so where p is close to one, which means that it measures target weakness in a known proportion and not transferability. The target-correct protocol removes that confound but admits samples the source itself cannot solve, and because those marginal subsets are highly vulnerable and asymmetrically sized, it pulls two directions toward each other by an amount that depends on the clean-accuracy gap. The both-correct protocol scores both directions on identical samples and is the only one of the four that supports paired inference, which is why we treat it as the primary estimate. The successful-source protocol answers a different question, how well an attack that already worked carries over, and its larger value reflects that conditioning on source success selects strong perturbations.

Across four protocols and three training runs, CNN-crafted examples transfer better to the ViT than the reverse *on both CIFAR datasets*, and on CIFAR-10 also under MI-FGSM. The decomposition places that direction on the source side: examples crafted on our ViT-Tiny transfer worse to every target, the external WideResNet included, and in the external pool this ViT-Tiny is the weakest robust source while a robust XCiT-S12, an attention model with convolutional components, is not. On near-matched SVHN models two protocols report a CNN advantage and two a ViT advantage, and the primary both-correct protocol gives -0.35 points with equivalence accepted (Section IV-D2). The direction is therefore a property of this unmatched pair: there the protocol changes how large the effect looks and whether equivalence can be claimed, and once the clean accuracies are close it can change the sign as well. The failure of a transformer-specific attack [45] to rescue the ViT direction is consistent with a source that is weak for reasons the attack does not reach.

The conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. In our corrected runs both favor the CNN, so the decomposition and the headline agree. An earlier checkpoint pair, trained before the leakage-free protocol, gave the same absolute ordering but a different division between the two factors (Section III-E), so a paper reporting only absolute accuracy would have seen nothing change, while a paper reporting the decomposition from a single run could have drawn a mechanistic conclusion that does not replicate.

The three readings that did not hold share a lesson about evidence. Sparsity measures [61] do not establish spatial locality: adversarially trained CNN gradients concentrate more mass on fewer components (Hoyer 0.4928 versus 0.4561, consistent in every run but not significant at the run level), yet four spatial descriptors show no difference in how those components are arranged. CLS attention entropy is unchanged under attack at $n = 1000$ while attention maps are displaced by a small, depth-increasing amount, and attention-shift results obtained with patch attacks [32], [33] should not be carried over to L_∞-bounded perturbations without measurement. TGR [45] costs 14.9 points of white-box strength and returns none of it at the target in our regime. Two behavioral differences do survive: the sparsity ordering, which adversarial training creates, together with an alignment difference in absolute cosine that comes with a near-zero signed mean, so that samples share sensitivity axes but not directions, unlike what a universal-perturbation claim [66] would need; and the depth profiles, concentrated in the last residual stage of the CNN and spread across depth in the ViT, with a measured ViT profile that depends on token aggregation.

B. REPORTING REQUIREMENTS

The measurements translate into requirements for cross-architecture transfer and robustness studies. They are ordered

by how far the corresponding choice moved a conclusion in this paper, and the first two matter most.

1) *State the conditioning set of every transfer number*, including sub-choices such as the strict or loose reading of successful-source, which alone moved our spread by 4.77 points (Section IV-B, Table 1).
2) *Do not present unconditioned transfer rates as architecture comparisons.* With per-sample records the target-correct rate can be computed exactly. Where only published raw rates and clean errors are available, Equation (6) bounds it, $r_{\text{cond}} \in [(r_{\text{raw}} - e)/(1-e), (r_{\text{raw}} - e\,p_{\text{lo}})/(1-e)]$, with p_{lo} the lowest premise to be expected in the regime: 0.984 on the adversarially trained targets we measured and 0.877 on standard-trained ones (Sections IV-B1 and IV-E).
3) For directional claims, prefer a protocol that scores both directions on identical samples, so that paired inference is available (Section IV-B).
4) Report the clean accuracy of every model and the clean-accuracy gap of every compared pair, since the identity term scales with that gap (Sections IV-D4 and IV-E).
5) Before attributing a direction to the target, separate source and target effects, for instance with a fixed third model (Section IV-C).
6) State the checkpoint-selection rule, which moved both absolute robust accuracy and the protocol spread by about two points on fixed trajectories (Section IV-F).
7) Report absolute robustness together with its factorization into clean accuracy and conditional susceptibility, and base mechanism claims on more than one training run: an earlier checkpoint pair of ours kept the absolute ordering but divided it differently between the two factors (Sections III-E and IV-A).
8) For layer-wise analyses, state the token aggregation and the tensor, and check attention claims on hundreds of samples, not on a figure batch (Section IV-G2).

For practitioners choosing between the two families under the matched mixed-loss recipe on this benchmark, the CNN gives higher absolute robust accuracy and the advantage holds under every view we measured. The WideResNet reference at 62.76% shows that a larger model trained on extra data with a stronger recipe [54] moves robustness further than the choice between these two families does.

C. LIMITATIONS

Every architectural conclusion here rests on one adversarially trained pair of small models, ResNet-18 and a ViT-Tiny trained from scratch on 32×32 inputs upsampled to 224×224, under one mixed-loss PGD adversarial fine-tuning recipe; ViT adversarial training is known to be recipe-sensitive on small datasets [56], [57], a ViT trained with the diffusion-generated data recipe of Wang et al. [67] has been reported to surpass WideResNets on CIFAR-10 [36], and the two models differ in parameter count (11.2M versus 5.7M). Only the parts of the 15-point CIFAR-10 range that come from the clean-accuracy difference should shrink for pairs matched in clean accuracy, as SVHN and the standard-trained pairs indicate, while the successful-source excess stays large in matched robust pairs; SVHN itself rests on two pairs, 50 epochs and no AutoAttack, and two runs cannot bound its run-to-run variation. The exact form of Equation (7) is validated only where p is close to one, which in our data means adversarially trained targets. The mechanism account is incomplete: the composition of marginal subsets covers three of the four protocols, and the successful-source protocol adds a second driver tied to the source's own robustness, whose effect on the spread we could not estimate reliably. Our registrations were committed before the corresponding measurements but not before training: the CIFAR-100 registration followed the start of training, and the fit pool first used excluded SVHN through a reassignment written after its result, so the SVHN-inclusive fit is the binding one. Every model is trained under L_∞ at $\epsilon = 8/255$, so the L_2 result probes the protocol and not L_2 robustness. We cannot isolate why our two checkpoint sets differ, because the corrected protocol changed the validation handling and the training run at once; the leakage ablation we ran on six full-budget trajectories did not detect an effect of leakage on checkpoint selection, but it answers a weaker question than the one posed, since the treatment used a single validation split and split identity and leakage are not separable in that design. Our TGR result, finally, comes from an adaptation of the original method and not a reproduction of its published setting, and the CIFAR-native ViT control calibrates the upsampling confound without providing a second matched pair.

D. FUTURE DIRECTIONS

The most direct extension is a calibration curve: architecture pairs with controlled clean-accuracy gaps, trained under one shared recipe, would turn the matched-pair prediction from a handful of points into a measured function and show whether the successful-source term depends on source robustness as the external pool suggests. A capacity-matched pair (ViT-Small against ResNet-50, say) would test whether the source effect, the sparsity ordering and the depth profiles belong to the families or to these particular small models. Re-scoring published transfer studies needs per-direction clean accuracies and, for an exact correction, per-sample records; studies that release them make the correction of Equation (6) a routine step.

VI. CONCLUSION

We asked how much the conditioning set of a transfer rate decides what a CNN-versus-ViT robustness comparison concludes, and answered it by varying only that set on fixed models and data. The conditioning set decides more than retraining does: on one adversarially trained pair it moves the measured asymmetry across a range several times wider than the across-run range of any single protocol.

Most of that range is predictable. The part between unconditioned and target-correct rates is the target's clean error

entering through the law of total probability, an exact correction where its premise has been measured close to one and an upper bound where it has not; the successful-source protocol answers a different question; and the two protocols that remain defensible still differ by a composition term that the marginal subsets of the two models account for. A transfer number is therefore interpretable only together with the set it was computed on.

What the asymmetry says about the architectures is narrower than its size suggests. Its direction comes from the examples crafted on this particular ViT-Tiny, which transfer worse to every target, and not from a weaker ViT target; the part of its range that the identity carries follows the clean-accuracy gap of the pair and shrinks for pairs matched in clean accuracy, in our own models and in an external pool alike, whereas the successful-source excess does not shrink, and the checkpoint-selection rule shifts the range by a few points. Absolute robustness favored the CNN under every attack, and three readings that architecture comparisons readily suggest did not survive direct measurement.

Our study covers three datasets and a single adversarially trained model pair, with standard-trained and external models used only for the protocol analysis, so the size of the effect for other pairs remains to be calibrated. Because every protocol estimate is recomputed from per-sample records, the comparison can be repeated on other models without new attacks, and the reporting requirements of Section V-B state what a reader needs to make that possible.

DATA AND CODE AVAILABILITY

The experiments in this study use the CIFAR-10 and CIFAR-100 datasets [13] and the SVHN dataset [62], all of which are publicly available. Pre-trained WideResNet-28-10 weights were obtained from the RobustBench benchmark [14], [54]. The analysis pipeline, the scripts that generate every table and figure in this paper, the per-sample evaluation records, the fixed validation-split indices and the seed lists are publicly available at `https://github.com/cagrisahin58/cnn-vit-robustness-protocols`. The per-sample records include the transfer runs of the clean-trained model pairs, of the eight-model RobustBench pool and of the last-checkpoint re-scoring, and the repository also holds a machine-read copy of Tables 1 and 2 of Mahmood et al. [8] together with the scripts that decompose and re-score them. Together these reproduce every number reported here, and because the four conditioning protocols are all computed from the per-sample records, the protocol comparison can be repeated without retraining. Trained checkpoints total about 40 GB and are available from the authors on request.

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